Wireless Technology 2025 May PYQ Answers

Q1. [5 Marks each]

a) Explain the difference between an Infrastructure-based Network and an Ad-hoc Network of WLAN.

b) Compare Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS).

c) Explain the role of SGSN and GGSN in GPRS.

d) Outline the method that supports mobility in CISCO Unified Wireless Network.

e) Compare IEEE 802.11 and IEEE 802.16.

Q2. [10 Marks each]

a) Draw and explain 4G network architecture and compare it with 3G.

b) What are the different types of GSM logical Channels? Explain in detail.

Q3. [10 Marks each]

a) Give the significance of WEP protocols. What are the features of WPA2?

b) Give the features of Cisco Unified Wireless Network and explain its architecture.

Q4. [10 Marks each]

a) What are the various challenges in WSN and explain the architecture of WSN?

b) What do you mean by massive MIMO? Compare 1G-5G mobile standards.

Q5. [10 Marks each]

a) Explain Zigbee protocol stack.

b) i) Compare VANET and MANET. ii) State the features of LoRaWAN.

Q6. [10 Marks each]

a) Draw and explain Bluetooth Architecture.

b) Describe the system architecture of 802.11.

Q1. [5 Marks each] - Answers

a. Explain the difference between an Infrastructure-based Network and an Ad-hoc Network of WLAN.

Basis	Infrastructure-based Network	Ad-hoc Network
Definition	A WLAN where devices communicate through a central device called an Access Point (AP).	A WLAN where devices communicate directly with each other without an Access Point.
Communication Method	Data passes through the Access Point.	Devices communicate peer-to-peer directly.
Central Control	Has centralized control through AP.	No centralized control.
Network Setup	Requires Access Point and proper setup.	Easy to set up, no AP required.
Coverage Area	Larger coverage area due to AP support.	Limited coverage depending on device range.
Security	More secure because AP can implement security mechanisms.	Less secure due to absence of central security control.
Scalability	Supports more users and devices.	Suitable for small number of devices only.
Example	Home/office Wi-Fi network using a router.	File sharing between laptops using Wi-Fi Direct.

b. Compare Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS).

Feature	Frequency Hopping Spread Spectrum (FHSS)	Direct Sequence Spread Spectrum (DSSS)
Definition	Signal hops between different frequencies rapidly	Signal is spread by multiplying with pseudo-noise code
Technique	Uses rapid frequency changes (hopping)	Uses spreading code to widen bandwidth
Interference Handling	Resistant to narrowband interference	Resistant to both narrowband and multipath interference
Bandwidth Requirement	Narrower bandwidth per hop	Requires wider bandwidth
Complexity	Simple design, less complex	More complex due to code synchronization
Data Rate	Lower data rates	Higher data rates
Applications	Bluetooth, military communication	Wi-Fi (IEEE 802.11b), GPS

c. Explain the role of SGSN and GGSN in GPRS.

Definition

In GPRS (General Packet Radio Service), **SGSN** and **GGSN** are the two main network nodes that handle packet-switched data services.

Key Points

- **SGSN (Serving GPRS Support Node):**
  - Manages packet data services for mobile users.
  - Handles mobility management (tracking location of MS).
  - Performs authentication and security checks.
  - Routes data packets between Mobile Station (MS) and GGSN.
  - Maintains subscriber information (with HLR/VLR support).
- **GGSN (Gateway GPRS Support Node):**
  - Acts as the interface between GPRS network and external packet data networks (e.g., Internet).
  - Converts GPRS packets into IP packets and vice versa.
  - Allocates IP addresses to mobile devices.
  - Provides access control and charging functions.
  - Ensures proper routing of data to external networks.

d. Outline the method that supports mobility in CISCO Unified Wireless Network.

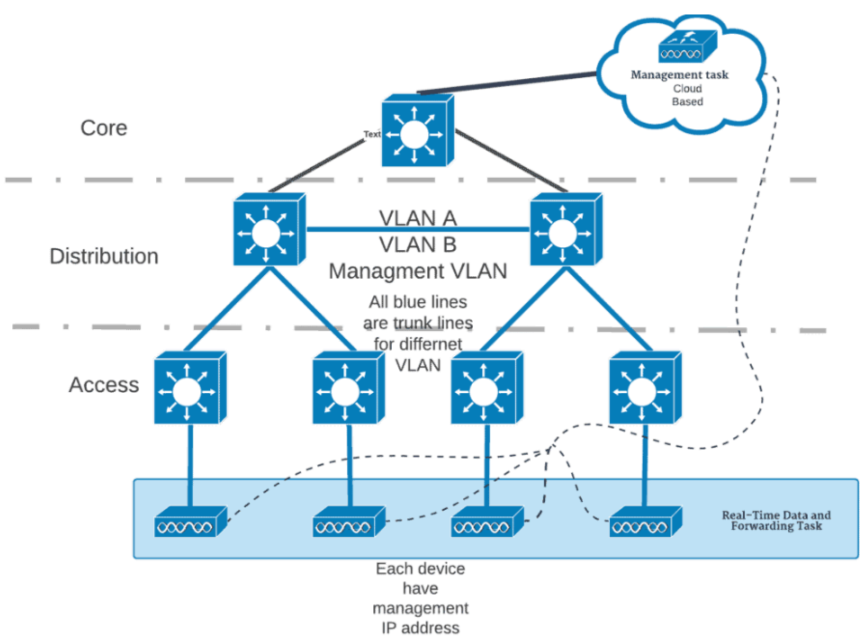
Cisco Unified Wireless Network supports mobility by allowing wireless users to move from one access point to another without losing connectivity.

Methods supporting mobility are:

1. **Layer 2 Roaming**
   - Allows a wireless client to move between Access Points within the same subnet.
   - IP address remains unchanged.
   - Provides seamless connectivity during movement.
2. **Layer 3 Roaming**

- Supports roaming between different subnets.
- User keeps the same IP address using mobility management.
- Prevents session interruption during movement.

3. **Inter-Controller Roaming**
   - Enables roaming between Access Points managed by different Wireless LAN Controllers (WLCs).
   - Controllers exchange client information for smooth handoff.
4. **Mobility Groups**
   - Multiple WLCs are grouped to share mobility information.
   - Helps clients roam easily across the network.
5. **Fast Secure Roaming (CCKM / 802.11r)**
   - Reduces delay during re-authentication when moving between APs.
   - Useful for voice and real-time applications.
6. **Access Point Handoff Mechanism**
   - Client automatically connects to the strongest AP signal while moving.
   - Ensures uninterrupted network access.



e. Compare IEEE 802.11 and IEEE 802.16.

Feature	IEEE 802.11 (Wi-Fi)	IEEE 802.16 (WiMAX)
Definition	Standard for Wireless LAN	Standard for Wireless Metropolitan Area Net
Coverage Range	Short range (up to ~100 m)	Long range (up to ~50 km)
Frequency Bands	2.4 GHz, 5 GHz	2–11 GHz (licensed/unlicensed)
Data Rate	Up to several hundred Mbps (e.g., 802.11ac/ax)	Up to 1 Gbps (fixed WiMAX)
Topology	Infrastructure (AP-based) or Ad-hoc	Point-to-multipoint, mesh
Applications	Home, office, hotspots	Broadband access, last-mile connectivity
Mobility Support	Limited mobility (basic roaming)	Better mobility support (802.16e for mobile)

Q2. [10 Marks each] - Answers

a. Draw and explain 4G network architecture and compare it with 3G.

Definition

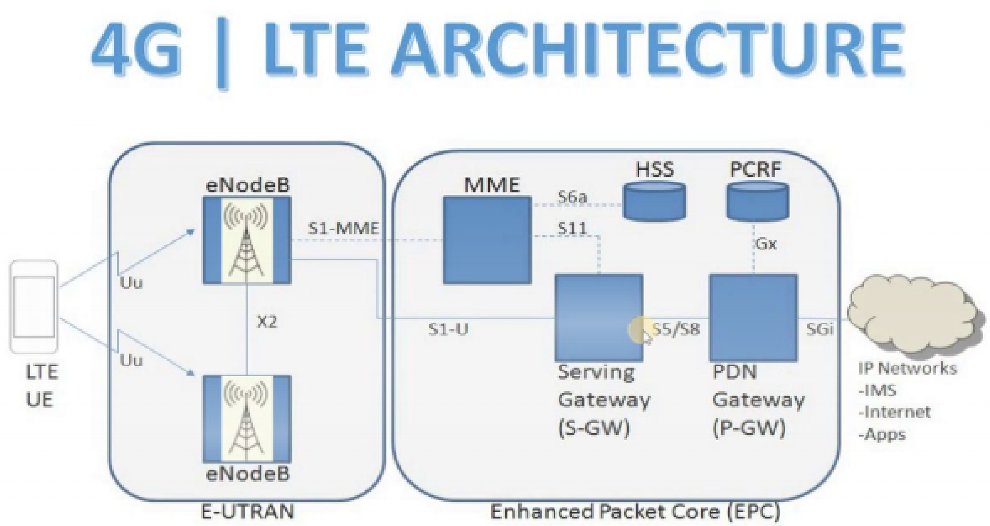
4G (Fourth Generation) is an all-IP based wireless communication system designed to provide very high-speed data, low latency, and improved multimedia services. It mainly uses **LTE (Long Term Evolution)** technology.

Architecture of 4G Network

1. **User Equipment (UE)**
    - Mobile device with SIM/USIM.
    - Connects to the network via radio interface.
2. **Evolved UMTS Terrestrial Radio Access Network (E-UTRAN)**
    - **eNodeB (Evolved Node B):**
        - Handles radio transmission/reception.
        - Manages scheduling, resource allocation, and mobility.
    - Interfaces:
        - **Uu:** UE ↔ eNodeB.
        - **X2:** Between eNodeBs for handover.

- **S1:** eNodeB ↔ EPC.

3. **Evolved Packet Core (EPC)**
    - **MME (Mobility Management Entity):** Authentication, mobility, signaling.
    - **SGW (Serving Gateway):** Routes user data, mobility anchor.
    - **PGW (Packet Gateway):** Connects to external IP networks, allocates IP addresses, QoS.
    - **HSS (Home Subscriber Server):** Stores user profiles and authentication data.
    - **PCRF (Policy and Charging Rules Function):** Controls QoS and charging policies.



Working of 4G Architecture

1. User Equipment (mobile) connects to **eNodeB**.
2. eNodeB communicates with **MME** for authentication.
3. User data passes through **S-GW** and **P-GW** to the internet.
4. HSS stores user identity and service information.

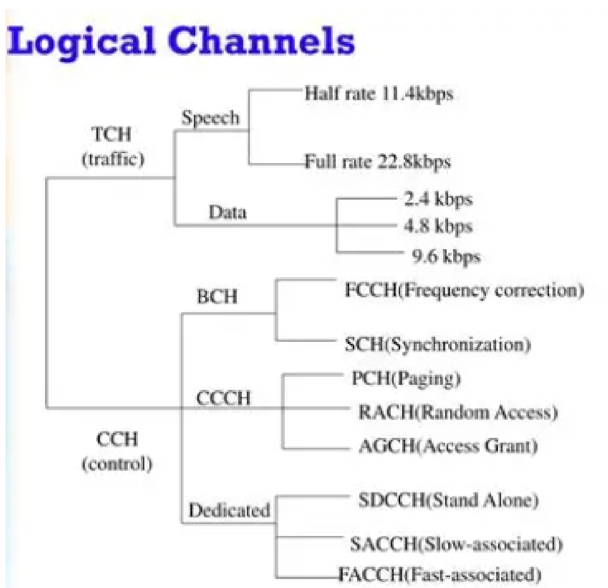
Feature	3G	4G
Generation	Third Generation	Fourth Generation
Technology	WCDMA, UMTS, CDMA2000	LTE, LTE-Advanced
Network Type	Circuit + Packet switched	Fully Packet switched (All-IP)
Data Speed	Up to 2 Mbps	100 Mbps to 1 Gbps
Latency	Higher	Very Low

Feature	3G	4G
Core Architecture	Complex layered architecture	Simplified EPC architecture
Voice Service	Circuit Switched Voice	VoIP / VoLTE
Multimedia Support	Moderate	High-quality HD multimedia
Spectral Efficiency	Lower	Higher
Handover	Slower	Faster and seamless

b. What are the different types of GSM logical Channels? Explain in detail.

Definition / Introduction

Logical channels in GSM define *what type of information* is carried over the physical radio channel. They are broadly divided into **Traffic Channels (TCH)** and **Control Channels (CCH)**.



Main Explanation

1. **Traffic Channels (TCH)**
    - Carry user data (voice or user information).
    - Types:
        - **Full-rate TCH (TCH/F):** 13 kbps voice.
        - **Half-rate TCH (TCH/H):** 6.5 kbps voice.
        - Can also carry circuit-switched data.
2. **Control Channels (CCH)**

- Used for signaling, system information, and call setup.
- Sub-types:

a) Broadcast Channels (BCH)

- **BCCH (Broadcast Control Channel):** Transmits cell identity, frequency list, system info.
- **FCCH (Frequency Correction Channel):** Helps mobiles synchronize frequency.
- **SCH (Synchronization Channel):** Provides frame synchronization and Base Station identity code.

b) Common Control Channels (CCCH)

- **PCH (Paging Channel):** Alerts mobile of incoming call/SMS.
- **RACH (Random Access Channel):** Used by mobile to request access.
- **AGCH (Access Grant Channel):** Grants access to requested resources.

c) Dedicated Control Channels (DCCH)

- **SDCCH (Standalone Dedicated Control Channel):** Used for call setup, SMS, location update.
- **SACCH (Slow Associated Control Channel):** Carries power control and timing advance info.
- **FACCH (Fast Associated Control Channel):** Used for urgent signaling (e.g., handover), temporarily replaces TCH.

Q3. [10 Marks each] - Answers

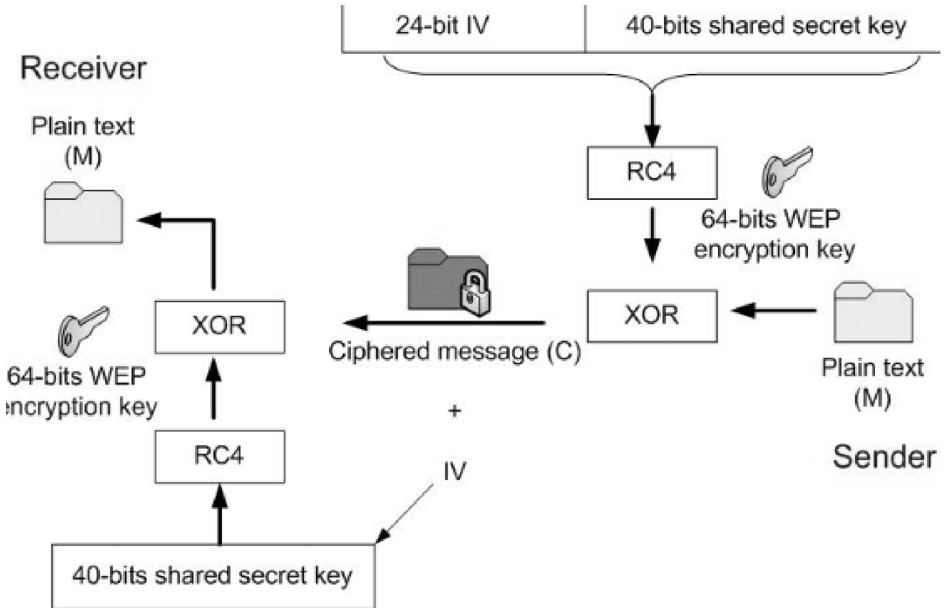
a. Give the significance of WEP protocols. What are the features of WPA2?

Significance of WEP Protocols

WEP (Wired Equivalent Privacy) was the first security protocol introduced for IEEE 802.11 wireless LANs, aimed at providing confidentiality similar to wired networks.

Explanation

- Ensured basic encryption using **RC4 stream cipher**.
- Used **40-bit or 104-bit keys** with a **24-bit Initialization Vector (IV)**.
- Provided **shared-key authentication** for access control.
- Offered **basic data integrity** using CRC-32 checksum.
- Helped establish the **foundation for wireless security standards**, though later found weak and easily cracked.



Features of WPA2

WPA2 (Wi-Fi Protected Access 2) is the successor to WPA, introduced by IEEE to overcome WEP's weaknesses. It is considered the most secure standard until WPA3.

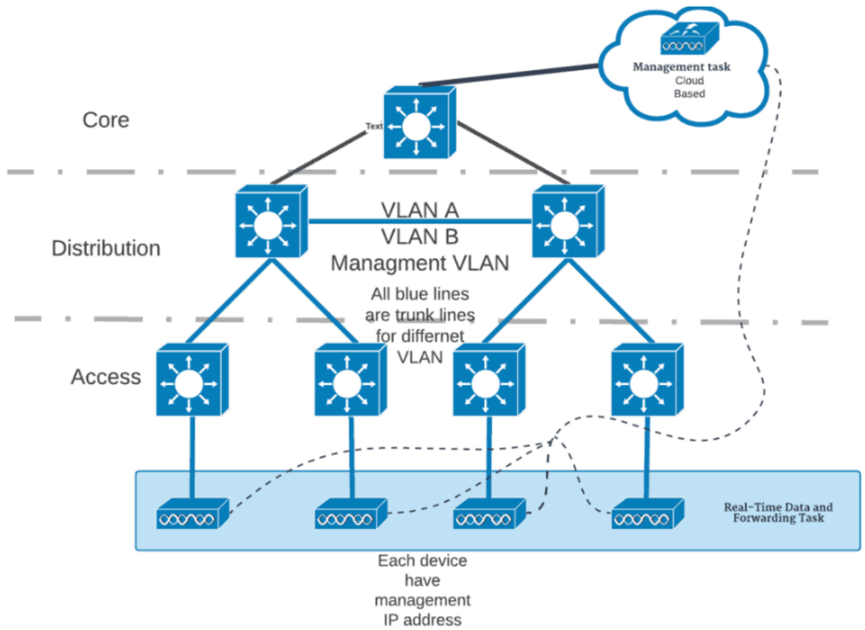
- **Strong Encryption:** Uses **AES (Advanced Encryption Standard)** with **CCMP** for confidentiality and integrity.
- **Robust Authentication:** Supports **IEEE 802.1X with EAP** for enterprise networks.
- **Data Integrity:** Prevents packet tampering with strong message integrity checks.
- **Dynamic Key Management:** Keys are generated and distributed securely, reducing risk of reuse.
- **High Security Standard:** Provides strong protection for both personal and enterprise WLANs.

b. Give the features of Cisco Unified Wireless Network and explain its architecture.

Features of CUWN

- **Centralized Management:** Wireless LAN Controllers (WLCs) manage multiple Access Points (APs).
- **Seamless Mobility:** Supports Layer 2 and Layer 3 roaming with mobility groups and anchors.
- **Enhanced Security:** Implements WPA2, 802.1X, and intrusion detection/prevention.

- **Scalability:** Can support thousands of APs and clients.
- **Quality of Service (QoS):** Prioritizes voice, video, and data traffic.
- **Reliability:** Redundancy and load balancing for high availability.



Architecture of CUWN

1. **Access Layer (Access Points)**
   - Lightweight APs connect users to the wireless network.
   - APs forward traffic to WLCs for centralized control.
2. **Distribution Layer (Wireless LAN Controllers)**
   - WLCs manage APs, handle authentication, mobility, and policy enforcement.
   - Provide seamless roaming and load balancing.
3. **Core Layer (Network Services)**
   - Integration with wired LAN, routing, and switching.
   - Connects to external networks and Internet.
4. **Network Management Layer**
   - Cisco Prime Infrastructure or similar tools for monitoring, configuration, and troubleshooting.

Key Components

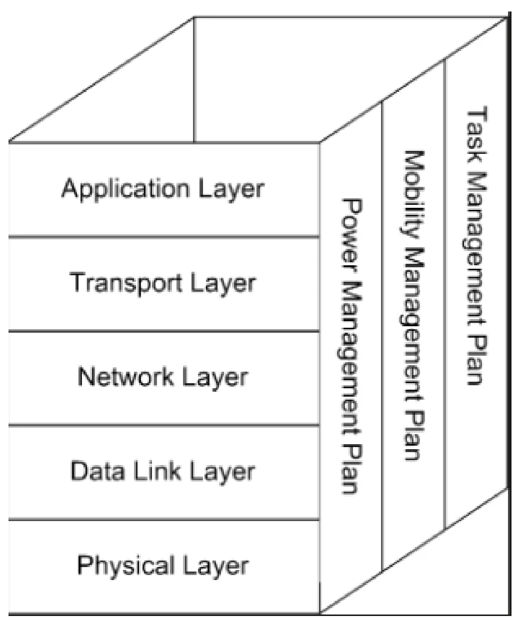
- **Access Points (APs):** Provide wireless coverage.
- **Wireless LAN Controller (WLC):** Centralized control of APs.
- **Mobility Services Engine (MSE):** Location tracking, context-aware services.
- **AAA Server (e.g., RADIUS):** Authentication and accounting.

Q4. [10 Marks each] - Answers

a. What are the various challenges in WSN and explain the architecture of WSN?

Challenges in Wireless Sensor Networks (WSN)

- **Energy Efficiency:** Sensor nodes have limited battery power; energy conservation is critical.
- **Scalability:** Networks may consist of hundreds/thousands of nodes, requiring efficient management.
- **Reliability:** Data delivery must remain accurate despite node failures or harsh environments.
- **Security:** Vulnerable to eavesdropping, tampering, and unauthorized access.
- **Data Aggregation:** Avoiding redundancy while maintaining accuracy in collected data.
- **Latency & QoS:** Meeting real-time requirements in applications like healthcare or military.



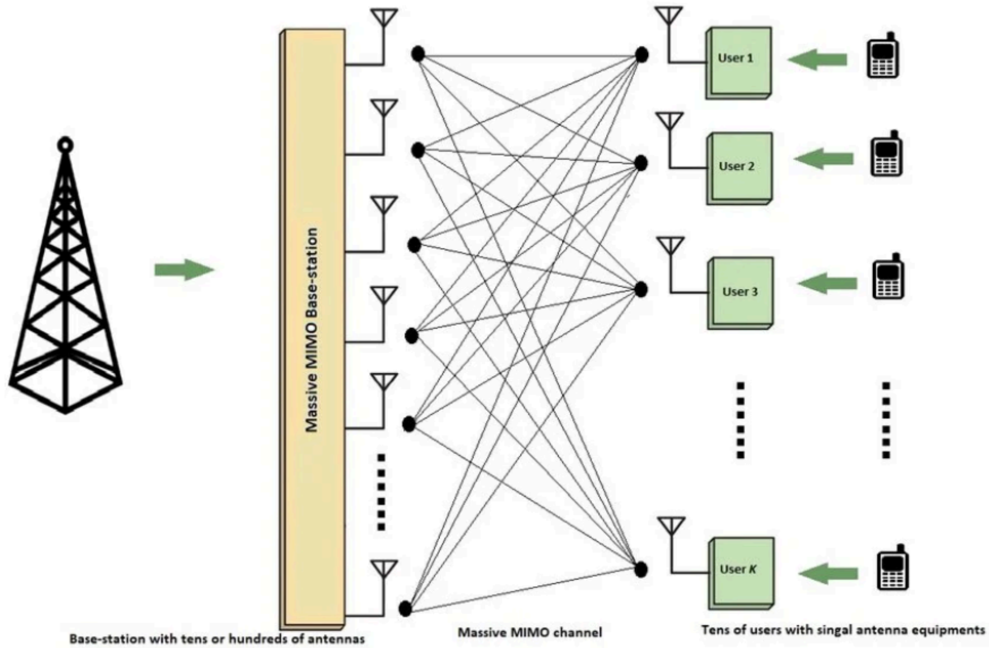
Architecture of WSN
WSN architecture is layered, similar to traditional networks, but optimized for low power and distributed sensing.

Layers of WSN Protocol Architecture

1. **Application Layer**
    - Provides application-specific services (e.g., monitoring, tracking).
    - Handles data formatting and user interface.
2. **Transport Layer**
    - Ensures reliable data delivery.
    - Uses congestion control and error recovery mechanisms.
3. **Network Layer**
    - Handles routing of data packets.
    - Focuses on energy-efficient, multi-hop communication.
4. **Data Link Layer (MAC Layer)**
    - Provides channel access, error detection, and frame synchronization.
    - Manages collisions and ensures efficient bandwidth use.
5. **Physical Layer**
    - Deals with modulation, transmission, and reception of signals.
    - Operates in ISM bands with low-power radios.

b. What do you mean by massive MIMO? Compare 1G-5G mobile standards.

Definition
Massive MIMO (Multiple Input Multiple Output) is an advanced wireless technology used in 5G networks where base stations are equipped with a very large number of antennas (tens to hundreds). It improves capacity, coverage, and reliability by serving multiple users simultaneously.



Working Principle

- In conventional MIMO, few antennas are used at transmitter and receiver.
- In Massive MIMO, a large antenna array is used at the base station.
- Multiple users can transmit and receive data at the same time.
- Uses **beamforming** to direct signals toward specific users.

Explanation

- **Large Antenna Arrays:** Base stations use many antennas to transmit/receive signals.
- **Spatial Multiplexing:** Multiple data streams are sent in parallel, increasing throughput.
- **Beamforming:** Directs signals precisely to users, reducing interference.
- **Energy Efficiency:** Focused transmission reduces power consumption.
- **Scalability:** Supports massive numbers of devices (IoT).
- **Key Role in 5G:** Enables ultra-high speed and low latency communication.

Generation	Technology & Access	Data Rate	Services	Limitations	Example Systems
1G (1980s)	Analog, FDMA	~2.4 kbps	Voice only	Poor quality, no security	AMPS

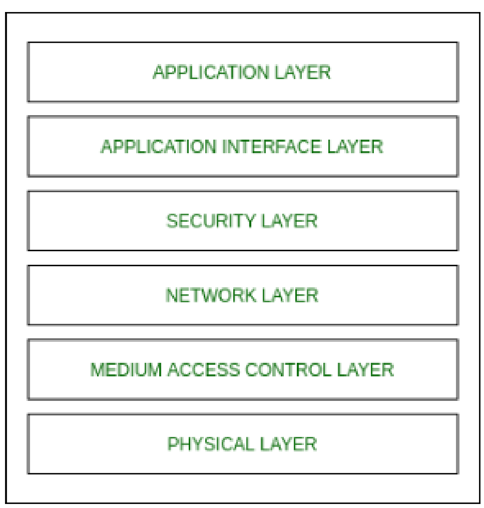
Generation	Technology & Access	Data Rate	Services	Limitations	Example Systems
2G (1990s)	Digital, TDMA/CDMA	~64 kbps	Voice, SMS, basic data	Limited speed	GSM, IS-95
3G (2000s)	WCDMA/UMTS, CDMA2000	~2 Mbps	Internet, video calls, multimedia	Higher latency, moderate speed	UMTS, CDMA2000
4G (2010s)	LTE, OFDMA	~100 Mbps (mobile), 1 Gbps (fixed)	HD video, VoIP, gaming, streaming	Requires all-IP, high infrastructure cost	LTE
5G (2020s)	OFDMA, Massive MIMO, mmWave	Up to 10 Gbps	Ultra-fast internet, IoT, AR/VR, autonomous systems	Complex deployment, high cost	5G NR

Q5. [10 Marks each] - Answers

a. Explain Zigbee protocol stack.

1. Definition

ZigBee is a wireless communication standard designed for low-power, low-data-rate applications such as home automation, industrial monitoring, and sensor networks. It is based on IEEE 802.15.4, which defines the lower layers, while ZigBee Alliance specifies the upper layers.



2. (Layers of ZigBee Protocol Stack)

1. **Physical Layer (PHY)**
    - Handles transmission/reception of data over radio.
    - Operates in **2.4 GHz ISM band** (global), 868 MHz (Europe), 915 MHz (North America).
    - Uses **O-QPSK modulation**.
    - Provides energy detection, link quality, and channel selection.
2. **MAC Layer (Medium Access Control)**
    - Manages access to the physical channel using **CSMA/CA**.
    - Provides frame validation, acknowledgements, and link quality monitoring.
    - Supports beaconing for synchronization.
3. **Network Layer**
    - Responsible for **network formation, addressing, and routing**.
    - Supports **star, tree, and mesh topologies**.
    - Uses routing protocols like **AODV (Ad hoc On-demand Distance Vector)**.
4. **Security Layer**
    - Provides **encryption, authentication, and integrity**.
    - Uses AES-128 encryption for secure communication.
5. **Application Support Sublayer (APS)**
    - Interfaces between network layer and application layer.
    - Manages binding tables, addressing, and data delivery.
6. **Application Layer**
    - Provides application profiles (e.g., home automation, health monitoring).
    - Defines device objects and services for end-user applications.

3. Example
A ZigBee sensor node in a smart home:

- **PHY/MAC:** Transmits temperature data wirelessly.
- **Network Layer:** Routes data via mesh topology to coordinator.
- **APS/Application:** Formats data for home automation profile.

b. i) Compare VANET and MANET. ii) State the features of LoRaWAN.

Feature	VANET (Vehicular Ad-hoc Network)	MANET (Mobile Ad-hoc Network)
Nodes	Vehicles equipped with wireless devices	Mobile devices (laptops, phones, sensors)
Mobility	Very high (fast-moving vehicles)	Moderate/low (pedestrian or portable devices)
Topology Change	Rapid and frequent	Relatively slower and less frequent
Communication Range	Larger (due to vehicle power and antennas)	Smaller, limited by device battery
Applications	Traffic management, safety, infotainment	Military, disaster recovery, personal networking
Challenges	High mobility, frequent disconnections	Limited resources, scalability, energy efficiency
Infrastructure Dependency	Often integrates with roadside units (RSUs)	Purely infrastructure-less, peer-to-peer

Definition

LoRaWAN (Long Range Wide Area Network): A low-power, wide-area networking protocol built on **LoRa modulation**. It connects battery-operated IoT devices over long distances with minimal energy consumption.

Key Points

- **Long Range Coverage:** 2–5 km in urban areas, 10–15 km in rural areas.
- **Low Power Consumption:** Enables devices to run on batteries for years.
- **Star-of-Stars Topology:** End devices connect to gateways, which forward data to a central network server.
- **Security:** End-to-end AES-128 encryption for data confidentiality and integrity.
- **Scalability:** Supports thousands of devices in a single network.
- **Adaptive Data Rate (ADR):** Optimizes data rate and transmission power for efficiency.
- **Applications:** Smart cities, agriculture, industrial IoT, environmental monitoring.

Q6. [10 Marks each] - Answers

a. Draw and explain Bluetooth Architecture.

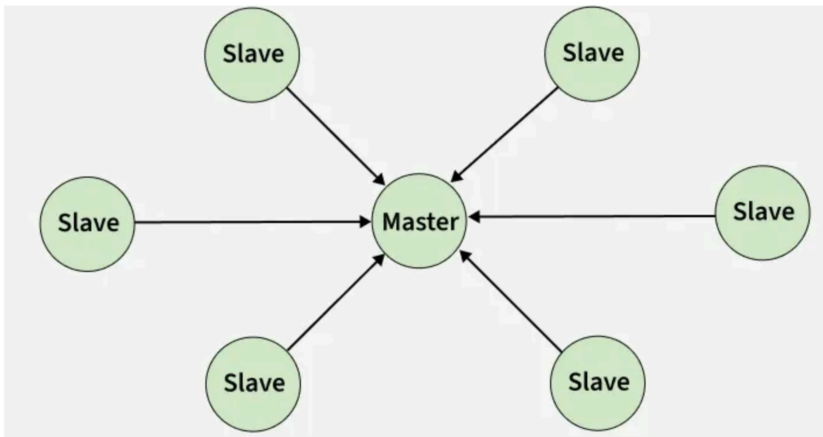
1. Definition

Bluetooth is a short-range wireless communication technology designed for connecting devices like phones, laptops, and peripherals. Its architecture defines how devices form networks and how the protocol stack enables communication.

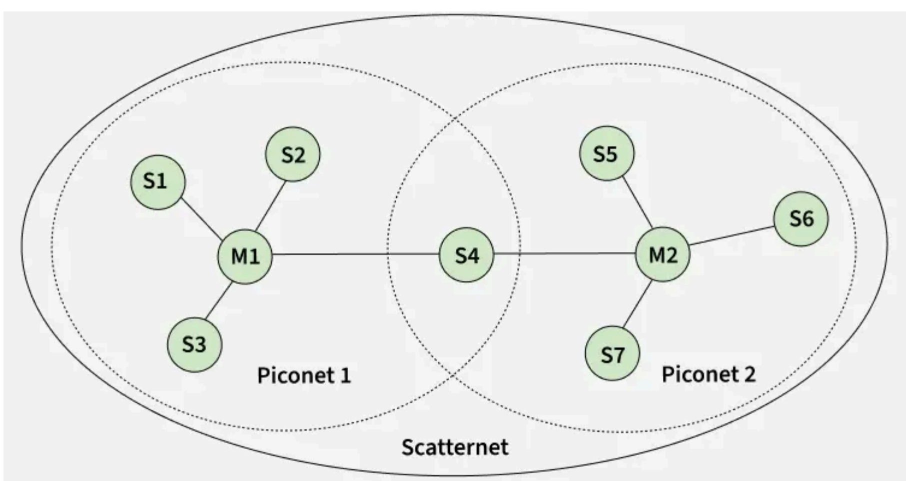
2. Main Explanation

A) Network Structures

- **Piconet:**
  - Basic Bluetooth network with one **master** and up to seven **active slaves**.
  - Master controls timing and communication.



- **Scatternet:**
  - Formed by interconnecting multiple piconets.
  - A device can act as master in one piconet and slave in another, enabling larger networks.



B) Bluetooth Protocol Stack Layers

1. **Radio Layer (Physical Layer):**

- Defines frequency (2.4 GHz ISM band), modulation, and transmit power.
- Handles conversion of digital data to RF signals.

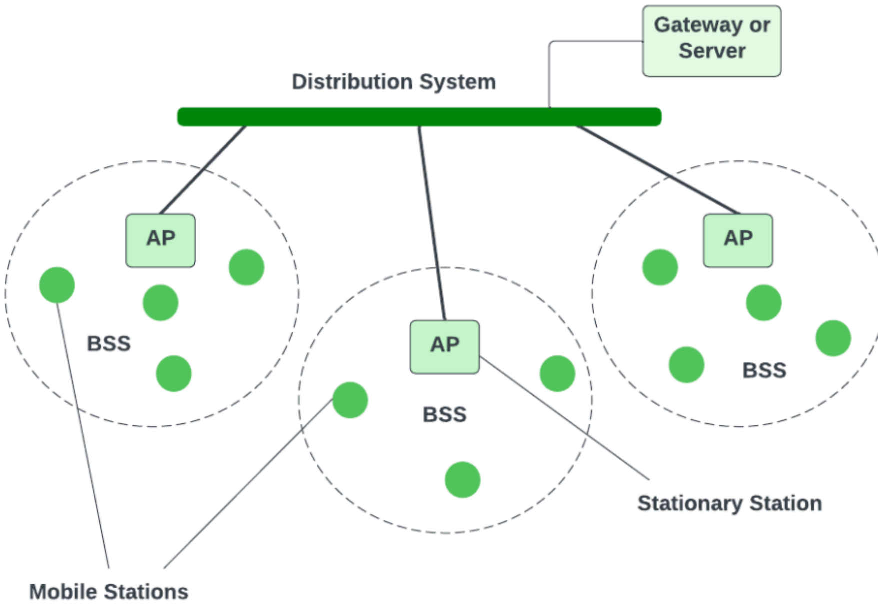
2. **Baseband Layer:**
   - Equivalent to MAC sublayer.
   - Manages device addressing, packet formats, and timing.
   - Establishes connections within a piconet.
3. **Link Manager Protocol (LMP):**
   - Handles link setup, authentication, encryption, and security.
   - Manages link states and termination.
4. **Logical Link Control and Adaptation Protocol (L2CAP):**
   - Provides communication between upper and lower layers.
   - Supports multiplexing, segmentation, and reassembly of packets.
5. **Service Discovery Protocol (SDP):**
   - Allows devices to discover available services on other Bluetooth devices.
6. **RFCOMM:**
   - Serial port emulation protocol.
   - Enables applications like file transfer and OBEX.
7. **Application Profiles:**
   - Define specific use cases (e.g., Hands-Free Profile, File Transfer Profile).

3. Example

- A Bluetooth headset connects to a smartphone:
  - **Piconet:** Phone = master, headset = slave.
  - **Protocol stack:** Radio → Baseband → LMP → L2CAP → RFCOMM → Hands-Free Profile.

b. Describe the system architecture of 802.11.

1. Definition
IEEE 802.11 is the standard for Wireless LAN (Wi-Fi). Its system architecture defines how stations (devices) connect through access points to form wireless networks, enabling communication with wired infrastructure.



ESS - Extended Service Set

Key Components:

1. **Station (STA):**
    - Any device with a wireless NIC (laptop, phone).
    - Communicates via radio interface.
2. **Access Point (AP):**
    - Connects wireless stations to the wired LAN.
    - Provides centralized control in infrastructure mode.
3. **Basic Service Set (BSS):**
    - A group of STAs controlled by one AP.
    - Smallest building block of WLAN.
4. **Extended Service Set (ESS):**
    - Multiple BSSs interconnected via a Distribution System (DS).
    - Provides seamless coverage across larger areas.
5. **Distribution System (DS):**
    - Backbone network connecting APs to each other and to the wired LAN.
    - Enables roaming between APs.

Wireless Technology 2025 December PYQ Answers

Q1. [5 Marks each]

a) Outline the method that supports mobility in CISCO Unified Wireless Network.

b) Explain the different features of VANET and E-VANET.

c) State the advantages of TDMA over FDMA.

d) Differentiate between FHSS and DSSS.

e) Write a note on UMTS security.

Q2. [10 Marks each]

a) Write short notes on: EDGE for 2.5G GSM and IS-136.

b) Draw and explain the GSM time slot hierarchy.

Q3. [10 Marks each]

a) Explain the architecture of Bluetooth. Explain active, hold, and park mode of operations in Bluetooth.

b) Draw and explain the architecture of Cisco UWN (Unified Wireless Network) with its features.

Q4. [10 Marks each]

a) Draw and explain MANET. What are the design challenges in MANET?

b) Draw and explain LTE network architecture in detail.

Q5. [10 Marks each]

a) Draw and explain wireless sensor network (WSN) communication architecture.

b) Explain 4G network architecture with its specifications.

Q6. [10 Marks each]

a) With the help of a neat diagram, explain the digital signature process.

b) Draw and explain the OFDMA technique for multiplexing.

Q1. [5 Marks each] - Answers

b. Explain the different features of VANET and E-VANET.

VANET (Vehicular Ad-hoc Network) is a wireless network where vehicles communicate with each other and roadside units. **E-VANET (Enhanced Vehicular Ad-hoc Network)** is an advanced form of VANET with improved intelligence, efficiency and safety.

Features of VANET

1. **Improves Road Safety:** Provides warnings about accidents, collisions, and hazardous road conditions.
2. **Traffic Management:** Helps monitor and control traffic flow, reducing congestion.
3. **Vehicle-to-Vehicle (V2V) Communication:** Vehicles exchange information like traffic and accident alerts.
4. **Vehicle-to-Infrastructure (V2I) Communication:** Communication with roadside units for traffic management.
5. **Supports Navigation and Route Guidance:** Provides real-time traffic updates and suggests better routes.
6. **Emergency and Accident Assistance:** Enables quick emergency alerts and faster response services.

Features of E-VANET

1. **Enhanced Communication Efficiency:** Provides faster and more reliable communication than VANET.

2. **Better Security and Privacy:** Includes stronger authentication and secure communication.
3. **Higher Quality of Service (QoS):** Supports low delay and reliable data transfer.
4. **Integration with Sensors and IoT:** Uses GPS, sensors, and smart devices for improved services.
5. **Advanced Safety and Emergency Services:** Supports emergency alerts, automated responses, and smart traffic control.

c. State the advantages of TDMA over FDMA.

Advantages of TDMA over FDMA

- **Better Spectrum Efficiency:** Multiple users share the same frequency, reducing wastage.
- **Higher Capacity:** Supports more users compared to FDMA with limited frequency bands.
- **Flexibility:** Time slots can be dynamically allocated based on demand.
- **Reduced Interference:** No adjacent channel interference as in FDMA.
- **Lower Hardware Cost:** Requires less filtering compared to FDMA systems.
- **Supports Digital Transmission:** Well suited for modern digital communication systems.

e. Write a note on UMTS security.

Definition

UMTS (Universal Mobile Telecommunications System) provides stronger security than GSM by ensuring confidentiality, integrity, and mutual authentication between user and network.

Key Points

- **Mutual Authentication:** Both user and network authenticate each other, preventing fake base station attacks.

- **Authentication Vectors (AVs):** Generated by the Authentication Center (AuC) containing RAND, XRES, CK, IK, and AUTN.
- **Encryption (Confidentiality):** User data and signaling are encrypted using Cipher Key (CK).
- **Integrity Protection:** Integrity Key (IK) ensures signaling messages are not modified, using MAC codes.
- **Temporary Identities:** Uses TMSI and other mechanisms to protect user identity.
- **Stronger Algorithms:** Employs KASUMI block cipher, more secure than GSM's A5/1.

Q2. [10 Marks each] - Answers

a. Write short notes on: EDGE for 2.5G GSM and IS-136.

i) EDGE for 2.5G GSM

- **Definition:** Enhanced Data rates for GSM Evolution (EDGE) is an enhancement over GSM/GPRS, often called 2.5G.
- **Key Points:**
    - Introduced **8-PSK modulation** to increase data rates.
    - Achieves speeds up to **384 kbps** (higher than GPRS ~80 kbps).
    - Backward compatible with GSM/GPRS networks.
    - Provides better support for multimedia services and Internet access.
- **Significance:** Served as a bridge between 2G and 3G, enabling faster mobile data before full 3G rollout.

ii) IS-136 (D-AMPS)

- **Definition:** IS-136, also known as Digital Advanced Mobile Phone System (D-AMPS), was a 2G digital cellular standard used mainly in North America.
- **Key Points:**

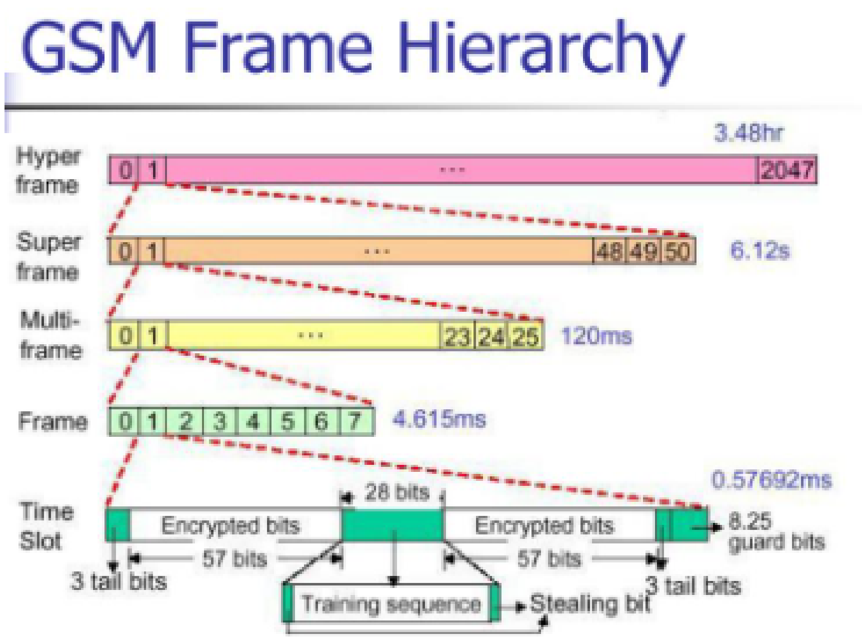
- Based on **TDMA (Time Division Multiple Access)**.
- Operated in **800 MHz and 1900 MHz bands**.
- Supported **voice, SMS, and limited data services**.
- Provided **better capacity and security** compared to analog AMPS.
- Allowed coexistence with analog systems during migration.

- **Significance:** Acted as a transitional digital standard before CDMA and GSM became dominant.

b. Draw and explain the GSM time slot hierarchy.

1. Definition

In GSM, radio resources are organized hierarchically into **frames, multiframes, superframes, and hyperframes**. This structure ensures efficient multiplexing of multiple users over the same frequency using TDMA.



2. Main Explanation

1. **Time Slot**
   - Basic unit of GSM transmission.

- Duration: **577 µs**.
- Each slot carries one burst (data or signaling).

2. **TDMA Frame**
   - Contains **8 time slots**.
   - Duration: **4.615 ms**.
   - Allows 8 users to share one frequency.
3. **Multiframe**
   - Two types:
     - **26-frame multiframe:** Used for traffic channels (TCH) + SACCH. Duration = **120 ms**.
     - **51-frame multiframe:** Used for control channels (BCCH, CCCH, etc.). Duration = **235.4 ms**.
4. **Superframe**
   - Contains **51 multiframes (traffic)** or **26 multiframes (control)**.
   - Duration = **6.12 s**.
5. **Hyperframe**
   - Contains **2048 superframes.**
   - Duration = **3 hours, 28 minutes, 53 seconds, 760 ms**.
   - Used for encryption key sequence management.

Q3. [10 Marks each] - Answers

a. Explain the architecture of Bluetooth. Explain active, hold, and park mode of operations in Bluetooth.

The answer for the architecture of Bluetooth is in May 2026(Q6) PYQ

Modes of Operation in Bluetooth

1. **Active Mode**

- Device actively participates in communication.
- Continuously listens and transmits data in assigned time slots.
- Consumes maximum power.

2. **Hold Mode**
    - Device temporarily suspends activity but remains synchronized with the master.
    - Saves power by not transmitting/receiving during hold period.
    - Useful when no data needs to be exchanged for a short time.
3. **Park Mode**
    - Device is synchronized but not actively participating.
    - Assigned a **parked address** instead of active address.
    - Can be reactivated by the master when needed.
    - Allows large numbers of devices to be connected but inactive.

Q4. [10 Marks each] - Answers

a. Draw and explain MANET. What are the design challenges in MANET?

1. Definition

A MANET is a self-configuring wireless network where mobile devices communicate directly without fixed infrastructure. Nodes act as both hosts and routers, enabling dynamic, multi-hop communication.

2. Architecture

- **Nodes (laptops, smartphones, sensors)** connect wirelessly.
- Each node can forward data for others (multi-hop).
- No base stations or routers are required.
- Topology changes dynamically as nodes move.

3. Working of MANET

1. Nodes join and form network dynamically.
2. Data is sent directly or through intermediate nodes.
3. Routing protocols establish and maintain routes.
4. If a node moves, route is updated automatically.

4. Design Challenges in MANET

1. **Limited Resources:** Nodes have restricted battery, bandwidth, and processing power.
2. **Scalability:** Performance degrades with large numbers of nodes.
3. **Security:** Vulnerable to eavesdropping, spoofing, and unauthorized access.

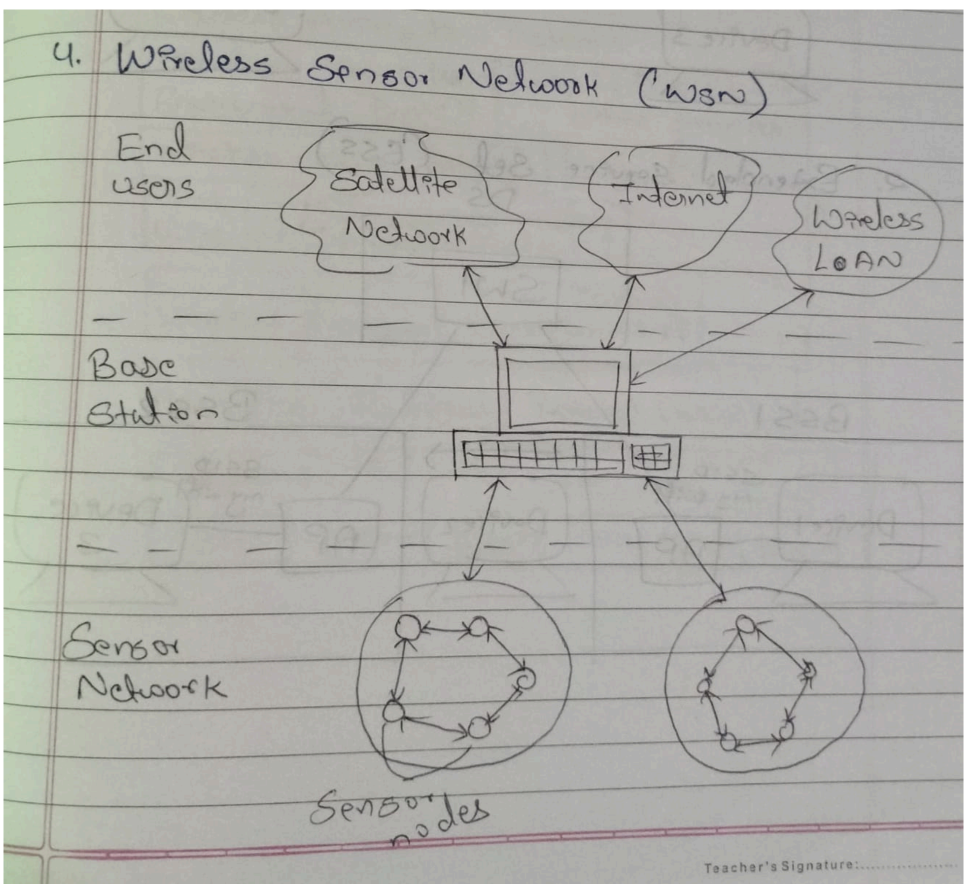
4. **Quality of Service (QoS):** Difficult to guarantee latency and reliability.
5. **Interference & Reliability:** Wireless links are prone to noise and fading.
6. **Energy Efficiency:** Power management is critical for mobile nodes.

Q5. [10 Marks each] - Answers

a. Draw and explain wireless sensor network (WSN) communication architecture.

1. Definition

Wireless Sensor Networks (WSNs) consist of distributed sensor nodes that sense, process, and transmit data wirelessly to a central base station or sink. The communication architecture defines how data flows from sensors to the end user through multiple layers and components.



2. Architecture Explanation

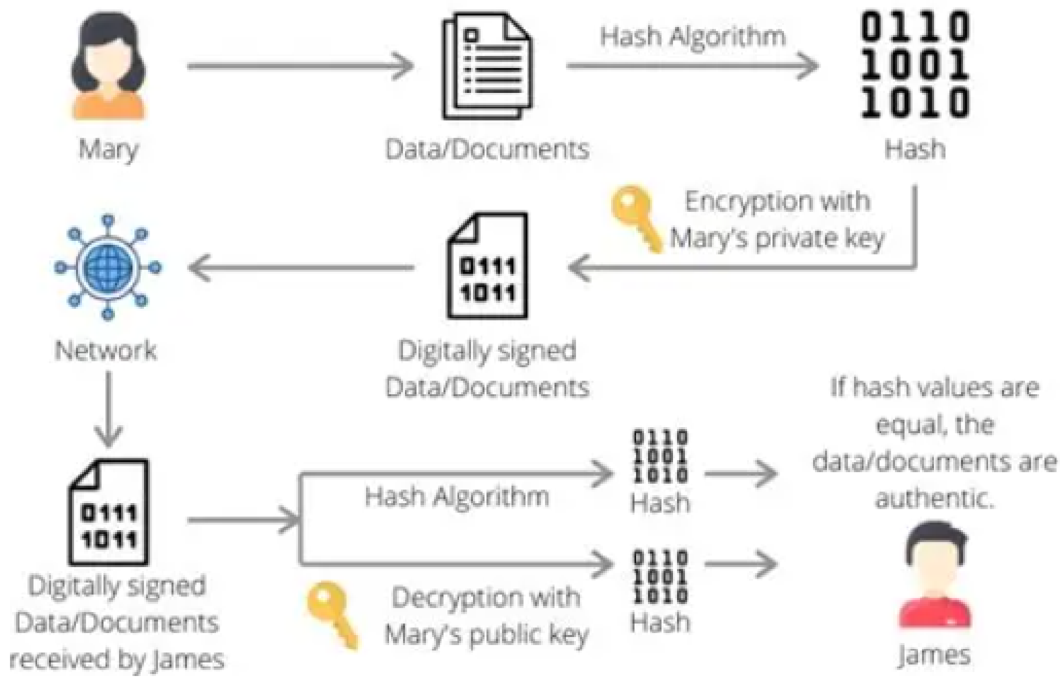
Key Components:

1. **Sensor Nodes**
    - Small devices with sensing, processing, and communication units.
    - Collect environmental data (temperature, pressure, motion, etc.).
2. **Cluster Heads (Optional in Hierarchical WSNs)**
    - Aggregate data from nearby sensor nodes.
    - Perform local processing to reduce redundancy.
3. **Communication Links**
    - Wireless multi-hop links between nodes.
    - Energy-efficient routing protocols used.
4. **Sink/Base Station (Gateway)**
    - Collects data from sensor nodes or cluster heads.
    - Acts as a bridge between WSN and external networks.
5. **External Network / End User**
    - Data is forwarded to servers, cloud, or applications.
    - End users access processed information for decision-making.

Q6. [10 Marks each] - Answers

a. With the help of a neat diagram, explain the digital signature process.

1. Definition
A digital signature is a cryptographic technique used to ensure **authenticity, integrity, and non-repudiation** of digital data. It verifies that the message was created by a known sender and was not altered during transmission.



2. Steps in Digital Signature Process

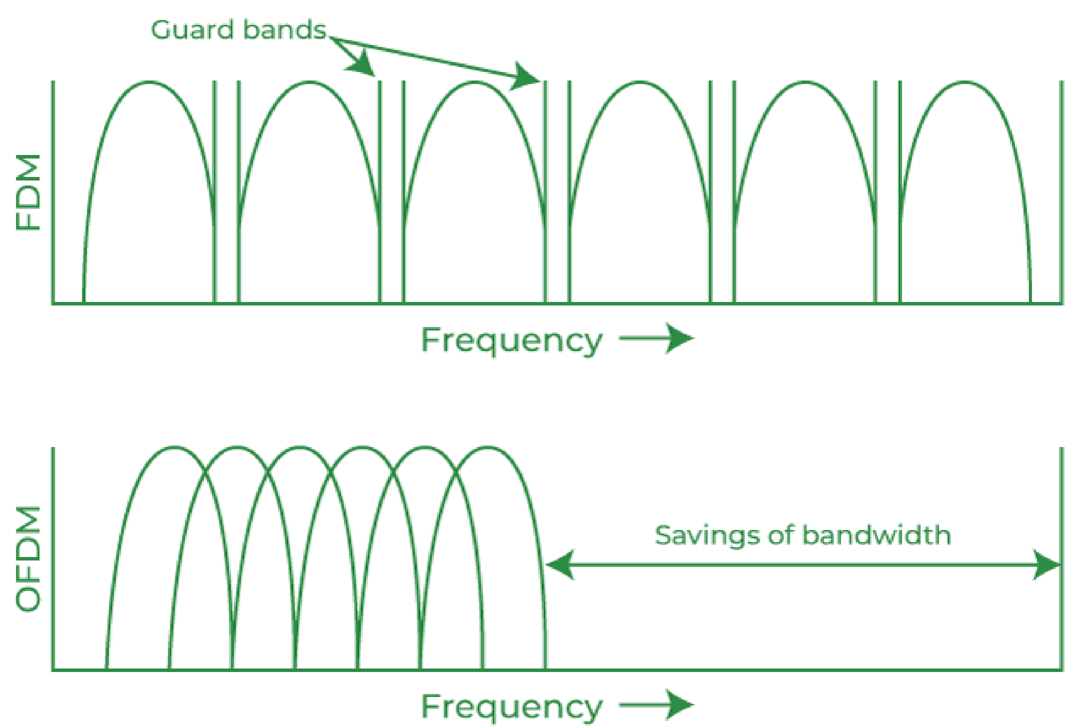
1. **Message Creation**
    - Sender prepares the original message/document.
2. **Hashing**
    - A hash function (e.g., SHA-256) generates a fixed-length hash value (message digest) from the original message.
    - Ensures integrity: even a small change in the message alters the hash.
3. **Encryption with Private Key**
    - The sender encrypts the hash using their **private key**.
    - This encrypted hash is the **digital signature**.
    - This encrypted hash is the **digital signature**.
4. **Transmission**
    - The sender sends the original message along with the digital signature to the receiver.
5. **Verification at Receiver Side**

- Receiver decrypts the signature using the sender's **public key** to obtain the hash.
- Receiver also generates a hash of the received message.
- If both hashes match → message is authentic and unaltered.

b. Draw and explain the OFDMA technique for multiplexing.

1. Definition

OFDMA (Orthogonal Frequency Division Multiple Access) is a **multi-user extension of OFDM**. It allows multiple users to share the same channel by assigning different subsets of orthogonal subcarriers to each user. This makes it highly efficient for modern wireless systems like **4G LTE and WiMAX**.



2. Working Principle

- **Spectrum Division:** The available bandwidth is divided into many narrowband orthogonal subcarriers.

- **Subcarrier Allocation:** Each user is allocated a group of subcarriers for transmission.
- **Orthogonality:** Subcarriers overlap in frequency but remain mathematically orthogonal, preventing interference.
- **Dynamic Resource Assignment:** Subcarriers can be allocated adaptively based on user demand, channel conditions, or QoS requirements.
- **Centralized Scheduling:** A base station assigns subcarriers and time slots to users, ensuring controlled access.

3. Advantages of OFDMA

- **Reduced Interference:** Orthogonality prevents overlap between users.
- **Scalability:** Efficiently supports large numbers of users.
- **Applications:** Used in LTE, WiMAX, and 5G systems.

Wireless Technology 2024 May PYQ Answers

Q1. [5 Marks each]

a) Give the features of VANET and MANET.
b) Compare Infrastructure-based Network and an Ad-hoc Network.
c) Outline the method that supports mobility in CISCO Unified Wireless Network.
d) Explain WEP and the security standard used.
e) Explain LoRaWAN.

Q2. [10 Marks each]

a) Explain the spread spectrum and briefly outline DSSS and FHSS.

b) Illustrate and explain GSM architecture with its interfaces.

Q3. [10 Marks each]

a) Outline the WSN architecture along with neat diagrams. State the various issues in WSN.
b) Illustrate the RF site survey process and its importance in the design process of designing a wireless network with Lightweight AP and WLC.

Q4. [10 Marks each]

a) State different features of Zigbee and explain its protocol stack.

b) Draw and explain Wi-Fi Architecture and Protocol Architecture.

Q5. [10 Marks each]

a) Draw and explain OFDMA with a neat diagram.

b) State the features of Wi-MAX and draw and explain the architecture of Wi-MAX.

Q6. [10 Marks each]

a) Explain UMTS and GSM security.

b) Compare and contrast 1G to 5G based on the technological differences and advancements. Write a short note on Massive MIMO.

Q1. [5 Marks each] - Answers

d. Explain WEP and the security standard used.

WEP (Wired Equivalent Privacy) is a security protocol used in wireless LANs to provide privacy and protect data transmission. It was designed to give wireless networks security similar to wired networks.

WEP (Wired Equivalent Privacy)

1. **Data Encryption:** WEP encrypts data before transmission to prevent unauthorized access.
2. **Uses RC4 Algorithm:** It uses the **RC4 stream cipher** for encryption.
3. **Secret Key Authentication:** Sender and receiver use a shared secret key for communication.
4. **Integrity Check:** Uses **CRC (Cyclic Redundancy Check)** to detect data modification.
5. **Key Sizes:** Commonly uses **64-bit** and **128-bit** keys.
6. **Weak Security:** WEP has vulnerabilities and can be easily cracked, so it is outdated.

Security Standards Used After WEP

1. **WPA (Wi-Fi Protected Access)**
   - Improved security over WEP.
   - Uses **TKIP** for better encryption.
2. **WPA2**

- Stronger security standard.
- Uses **AES (Advanced Encryption Standard)** encryption.

3. **WPA3**
   - Latest and more secure standard.
   - Provides stronger authentication and protection.

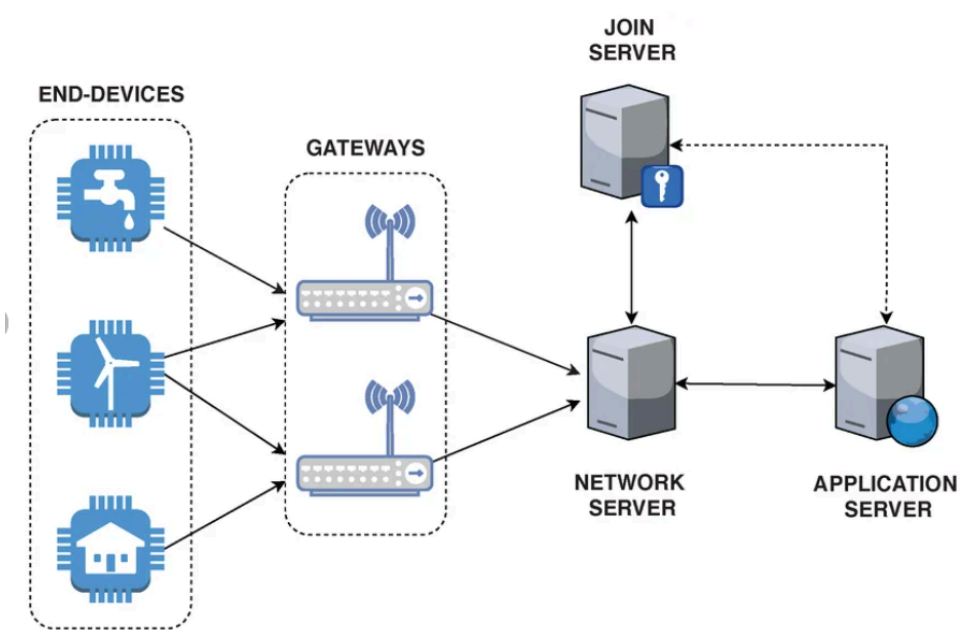
Example:

Early Wi-Fi networks used **WEP**, while modern networks use **WPA2 or WPA3** for better security.

e. Explain LoRaWAN.

Definition

LoRaWAN (Long Range Wide Area Network): A low-power, wide-area networking protocol built on **LoRa modulation**. It connects battery-operated IoT devices over long distances with minimal energy consumption.



Architecture of a LoRaWAN Network.

Key Points

- **Based on LoRa:** LoRa provides the physical layer (Chirp Spread Spectrum modulation), while LoRaWAN defines the communication protocol and system architecture.
- **Long Range:** Coverage up to **10–15 km in rural areas** and **2–5 km in urban areas**; even hundreds of km via satellite gateways.
- **Low Power:** Devices can last **years on a single battery**, optimized for IoT sensors.
- **Architecture:**
  - **End Devices (Nodes):** Sensors/actuators sending data.
  - **Gateways:** Forward LoRa signals to the network server.
  - **Network Server:** Manages devices, routing, and security.
  - **Application Server:** Processes data for end-user applications.
- **Security:** Provides end-to-end encryption using AES keys.
- **Applications:** Smart cities (parking, lighting), agriculture (soil monitoring), industrial IoT, and environmental monitoring.

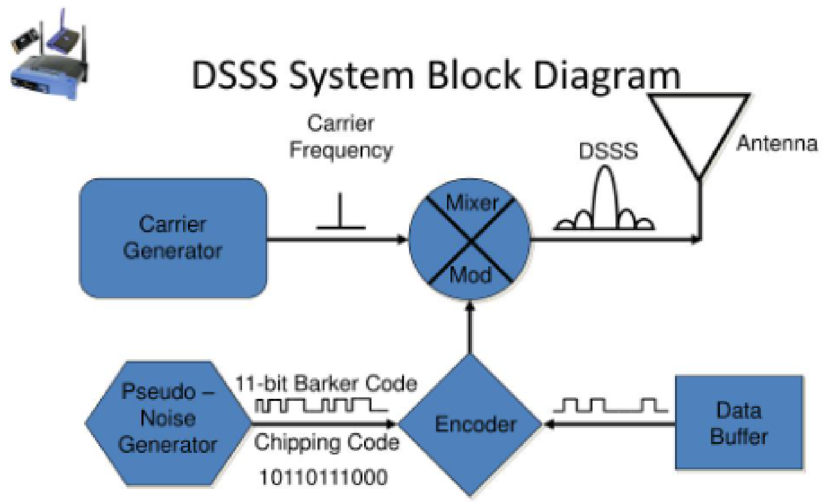
Q2. [10 Marks each] - Answers

a. Explain the spread spectrum and briefly outline DSSS and FHSS.

1. Spread Spectrum –

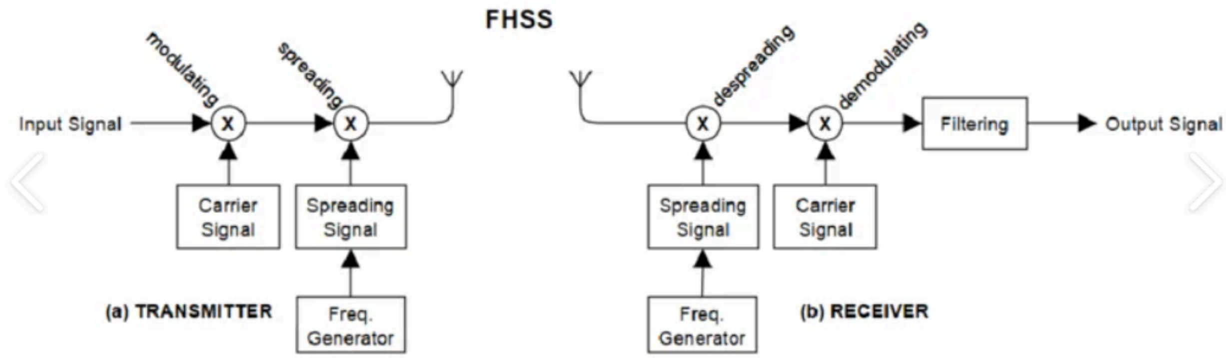
- **Definition:** A communication method where signals occupy a bandwidth much wider than the minimum required.
- **Purpose:**
  - Makes signals harder to intercept or jam.
  - Provides robustness against interference and multipath fading.
  - Improves security by using pseudo-random codes known only to sender and receiver.

- **Techniques:** DSSS and FHSS are the two primary approaches.



2. Direct Sequence Spread Spectrum (DSSS)

- **Process:**
  - Each data bit is multiplied by a high-rate pseudo-noise (PN) code called "chips."
  - This spreads the signal across a wider frequency band.
- **Advantages:**
  - Provides higher data rates (e.g., up to 11 Mbps in WLAN).
  - Easier to implement with less complex equipment.
- **Applications:**
  - IEEE 802.11b Wi-Fi, GPS, CDMA cellular systems.



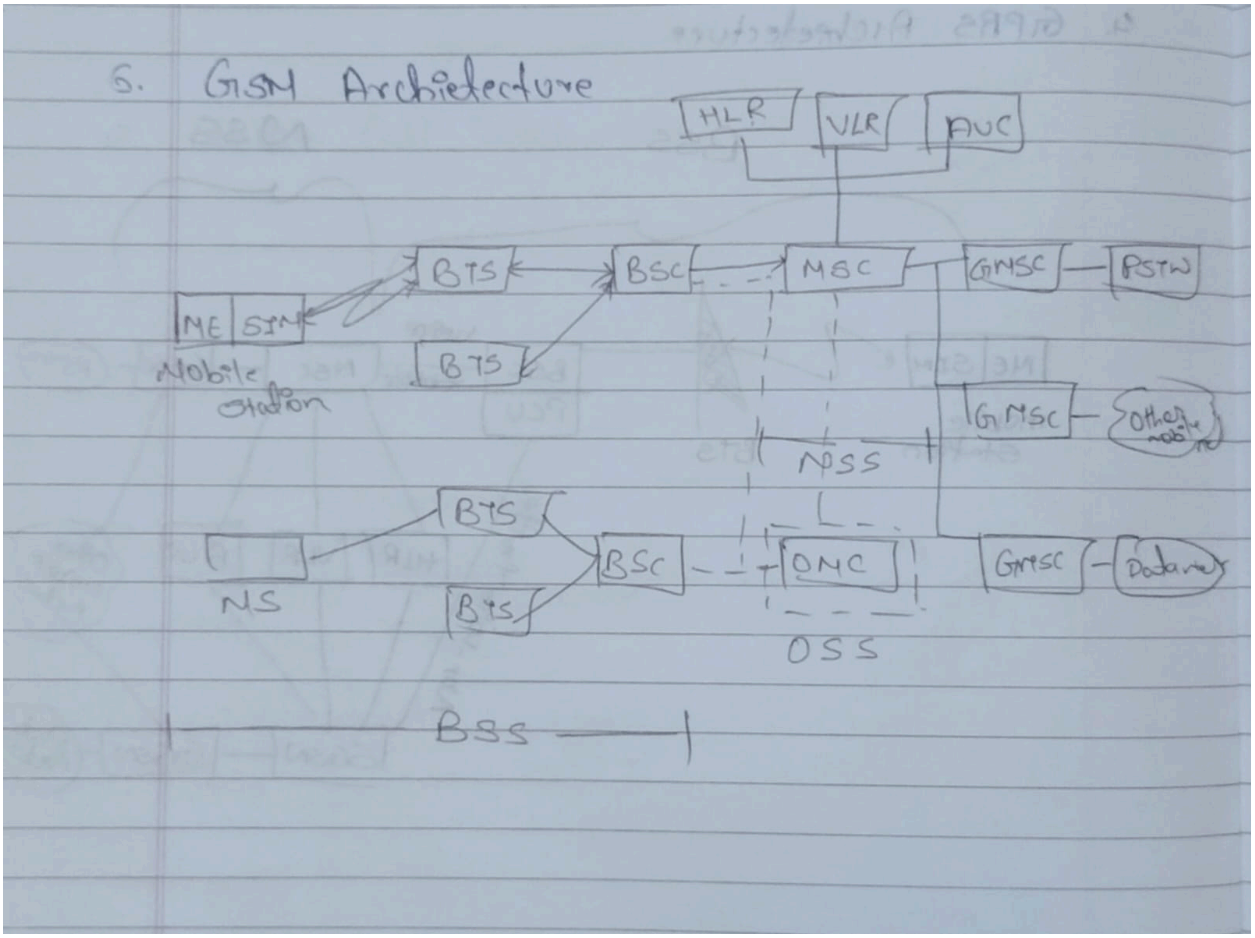
3. Frequency Hopping Spread Spectrum (FHSS)

- **Process:**
    - The carrier frequency rapidly hops between different channels according to a pseudo-random sequence known to both transmitter and receiver.
    - Each hop lasts for a short duration before moving to the next frequency.
- **Advantages:**
    - Very robust against interference and jamming.
    - Effective in point-to-multipoint scenarios.
- **Limitations:**
    - Lower data rates (around 3 Mbps compared to DSSS's 11 Mbps).
- **Applications:**
    - Bluetooth, military communications, walkie-talkies.

b. Illustrate and explain GSM architecture with its interfaces.

1. Definition

GSM (Global System for Mobile Communication) is a 2G digital cellular system. Its architecture is divided into subsystems with well-defined interfaces to ensure modularity, scalability, and interoperability.



2. GSM System Architecture

A) Mobile Station (MS)

- User’s handset + SIM card.
- Provides air interface for voice, SMS, and data.
- Interfaces: **Um** (radio interface between MS and BTS).

B) Base Station Subsystem (BSS)

- **BTS (Base Transceiver Station):** Handles radio transmission/reception with MS.
- **BSC (Base Station Controller):** Controls multiple BTSs, manages handovers, allocates radio channels.
- Interfaces:
  - **Um:** MS ↔ BTS.

    - **Abis:** BTS ↔ BSC.

C) Network Switching Subsystem (NSS)

- **MSC (Mobile Switching Center):** Central switching for calls and mobility.
- **HLR (Home Location Register):** Permanent subscriber database.
- **VLR (Visitor Location Register):** Temporary data for roaming users.
- **AuC (Authentication Center):** Provides security by verifying SIM identity.
- **EIR (Equipment Identity Register):** Tracks valid/invalid devices.
- Interfaces:
    - **A:** BSC ↔ MSC.
    - **MAP (Mobile Application Part):** MSC ↔ HLR/VLR/AuC/EIR.

D) Operation Support Subsystem (OSS)

- Provides network monitoring, maintenance, and management.
- Interfaces: **O&M (Operations & Maintenance)** with BSS/NSS.

Q3. [10 Marks each] - Answers

a. Illustrate the RF site survey process and its importance in the design process of designing a wireless network with Lightweight AP and WLC.

1. Definition
An RF site survey is the process of analyzing and measuring radio frequency behavior within a physical environment before deploying a wireless network. It ensures optimal placement of **Lightweight Access Points (APs)** and effective management by the **Wireless LAN Controller (WLC)**.

2. Steps in RF Site Survey

1. **Pre-Survey Planning**
    - Define coverage areas, capacity requirements, and applications (voice, video, data).

- Identify obstacles (walls, furniture, machinery).

2. **On-Site Measurement**
    - Use spectrum analyzers or survey tools to measure signal strength, interference, and noise levels.
    - Walk through the site to record RF propagation.
3. **Access Point Placement**
    - Determine optimal AP locations for maximum coverage and minimal overlap.
    - Ensure redundancy for seamless roaming.
4. **Channel & Power Planning**
    - Assign non-overlapping channels to APs.
    - Adjust transmit power to balance coverage and avoid interference.
5. **Validation Survey**
    - Test the deployed APs with WLC integration.
    - Verify coverage, throughput, and roaming performance.

3. Importance in CUWN (Lightweight AP + WLC Design)

- **Optimal Coverage:** Ensures no dead zones in the wireless network.
- **Capacity Planning:** Supports required number of users and applications.
- **Interference Mitigation:** Identifies and reduces co-channel and adjacent channel interference.
- **Seamless Mobility:** Guarantees smooth roaming between APs under WLC control.
- **Cost Efficiency:** Prevents over-deployment or under-deployment of APs.

Q4. [10 Marks each] - Answers

b. State different features of Zigbee and explain its protocol stack.

Key Features

- **Low Power Consumption:** Optimized for battery-operated devices, enabling years of operation.
- **Low Data Rate:** Supports up to **250 kbps**, suitable for sensor and control applications.
- **Network Topologies:** Supports **star, tree, and mesh** structures for flexibility.
- **High Scalability:** Can connect **thousands of devices** in a single network.
- **Security:** Provides AES-128 encryption for secure communication.
- **Cost-effective:** Simple design, low hardware cost, ideal for IoT deployments.
- **Applications:** Home automation, industrial monitoring, smart energy, healthcare.

The answer for Zigbee Protocol stack is in 2025 May (Q5→a)

Q5. [10 Marks each] - Answers

b. State the features of Wi-MAX and draw and explain the architecture of Wi-MAX.

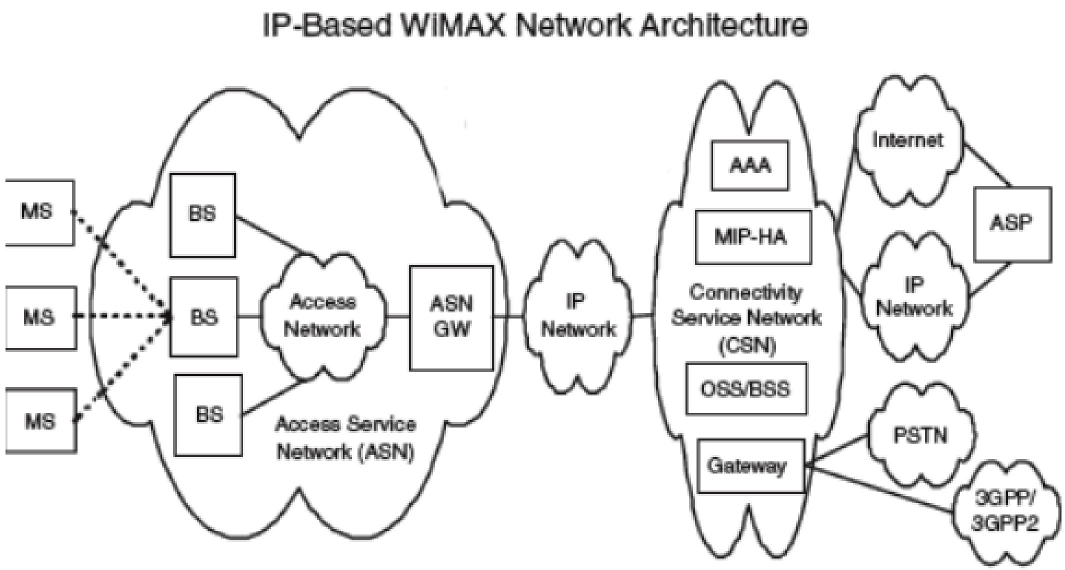
Definition

Wi-MAX (Worldwide Interoperability for Microwave Access): A wireless broadband technology based on IEEE 802.16 standard, designed to provide high-speed internet access over long distances.

Key Features

- **High Data Rates:** Supports up to **70 Mbps** depending on channel bandwidth.
- **Long Range Coverage:** Provides connectivity up to **30–50 km** in line-of-sight conditions.

- **Scalability:** Flexible architecture supporting point-to-point and point-to-multipoint communication.
- **Quality of Service (QoS):** Ensures priority handling for voice, video, and data traffic.
- **Security:** Uses strong encryption (AES, DES) and authentication mechanisms.
- **Cost-effective Deployment:** Reduces need for wired infrastructure, suitable for rural and urban areas.



Wi-MAX Architecture

1. Subscriber Station (SS):

- End-user device (CPE, modem, or mobile station).
- Connects wirelessly to the base station.

2. Base Station (BS):

- Provides wireless connectivity to multiple subscriber stations.
- Handles radio transmission, resource allocation, and QoS.

3. Access Service Network (ASN):

- Collection of base stations and gateways.

- Manages radio resources, mobility, and handovers.

4. Connectivity Service Network (CSN):

- Provides IP connectivity, authentication, billing, and roaming.
- Interfaces with external networks (Internet, PSTN, corporate).

5. External Networks:

- Internet, corporate intranets, or other service provider networks.

Q6. [10 Marks each] - Answers

a. Explain UMTS and GSM security.

1. GSM Security
GSM (2G) introduced basic security features but had several limitations.

- **Authentication:**
  - Based on a challenge-response mechanism.
  - Network sends a random number (RAND) to the SIM.
  - SIM computes response using secret key (Ki) and sends back SRES.
  - Only the **network authenticates the user**, not vice versa.
- **Confidentiality (Encryption):**
  - Uses ciphering algorithms (A5/1, A5/2) with session key (Kc).
  - Protects user data and signaling over the air interface.
- **Temporary Identity:**
  - Uses TMSI (Temporary Mobile Subscriber Identity) to hide IMSI.
- **Limitations:**
  - No mutual authentication (network not verified).
  - Weak encryption algorithms (A5/1 cracked).
  - Vulnerable to fake base station attacks.

2. UMTS Security (3G)
UMTS improved security significantly compared to GSM.

- **Mutual Authentication:**
  - Both user and network authenticate each other.
  - Prevents fake base station attacks.
- **Authentication Vectors (AVs):**
  - Generated by Authentication Center (AuC).
  - Includes RAND, XRES (expected response), CK (cipher key), IK (integrity key), AUTN (authentication token).
- **Encryption (Confidentiality):**
  - Uses CK to encrypt user data and signaling.
  - Stronger algorithms (KASUMI block cipher).
- **Integrity Protection:**
  - IK ensures signaling messages are not modified.
  - Provides Message Authentication Codes (MAC).
- **Temporary Identities:**
  - Uses TMSI and other mechanisms to protect user identity.
- **Advantages over GSM:**
  - Stronger algorithms.
  - Integrity protection added.
  - Mutual authentication ensures higher trust.

Wireless Technology 2024 December PYQ Answers

Q1. [5 Marks each]

a) How does OFDMA handle synchronization and interference between users?

b) List and explain the core components of the UMTS network.

c) Explain the difference between an Ad-hoc network and an infrastructure-based network.

d) What role does VANET play in intelligent transportation systems (ITS)?

e) Explain the function of a Subscriber Identity Module (SIM) in GSM.

Q2. [10 Marks each]

a) What is Frequency Hopping Spread Spectrum (FHSS), and how does it improve communication security?

b) Explain how LoRaWAN enables long-range, low-power communication by using spread spectrum techniques (Chirp Spread Spectrum).

Q3. [10 Marks each]

a) Explain the architecture and working of 4G.

b) How does Wi-Fi handle multiple users and avoid collisions in the network?

Q4. [10 Marks each]

a) Explain the architecture and Features of a WiMAX network.

b) Explain the pairing and security mechanisms used in Bluetooth communication.

Q5. [10 Marks each]

a) What is mesh networking in ZigBee, and how does it enhance the network's reliability?

b) What are the primary security mechanisms employed in GSM to protect user data and privacy?

Q6. [10 Marks each]

a) Discuss the flaws in WEP's initialization vector (IV) implementation and how it leads to key reuse and susceptibility to attacks.

b) How does Cisco implement security in its Unified Wireless Network?

Q1. [5 Marks each] - Answers

a. How does OFDMA handle synchronization and interference between users?

OFDMA (Orthogonal Frequency Division Multiple Access) allows multiple users to share the same channel by assigning different orthogonal subcarriers. It manages synchronization and interference using the following methods:

Synchronization Handling

1. **Orthogonal Subcarriers:** Users are assigned separate orthogonal subcarriers, preventing overlap.
2. **Time Synchronization:** Users are synchronized so transmissions arrive at the correct time.
3. **Cyclic Prefix (CP):** A cyclic prefix is added to reduce inter-symbol interference caused by delay.
4. **Frequency Synchronization:** Ensures carrier frequencies remain aligned to maintain orthogonality.

Interference Handling

1. **Subcarrier Allocation:** Different users get different subcarriers, reducing user-to-user interference.

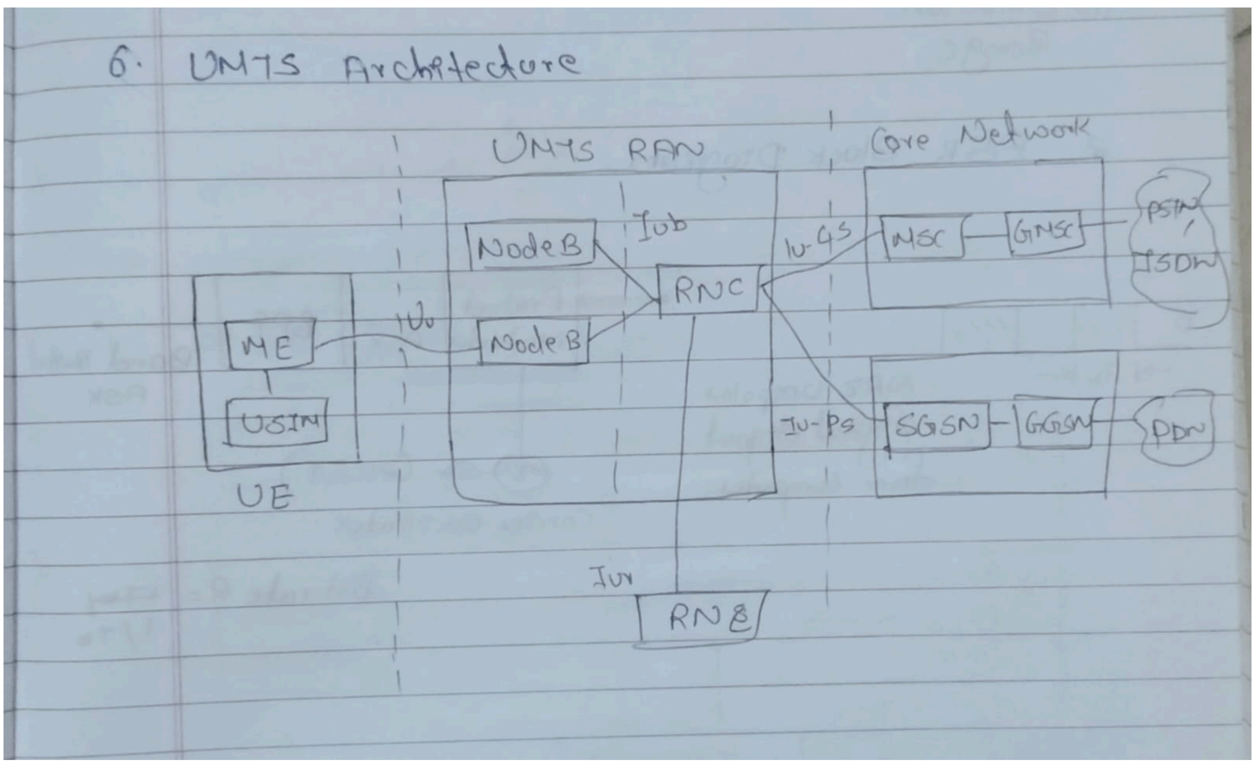
2. **Guard Bands / Orthogonality:** Orthogonality between subcarriers minimizes inter-carrier interference.
3. **Power Control and Scheduling:** The system controls transmission power and allocates resources efficiently to reduce interference.

Example:

In 4G/5G systems, OFDMA assigns different subcarriers to different users while synchronization and cyclic prefix prevent interference.

b. List and explain the core components of the UMTS network.

UMTS (Universal Mobile Telecommunications System) consists of major components that work together to provide mobile communication services.



Core Components

1. **User Equipment (UE)**
   - It is the mobile device used by the subscriber.

- Includes the handset and **USIM** for user identity and authentication.

2. **Node B**
   - It acts like a base station in UMTS.
   - Handles radio communication between user equipment and the network.
3. **Radio Network Controller (RNC)**
   - Controls multiple Node Bs.
   - Manages radio resources, handover, and connection control.
4. **UTRAN (UMTS Terrestrial Radio Access Network)**
   - Consists of **Node B and RNC**.
   - Provides radio access between users and the core network.
5. **Core Network (CN)**
   - Responsible for switching, routing, mobility management, and service delivery.
6. **MSC (Mobile Switching Center)**
   - Handles voice calls, call routing, and mobility management.
7. **SGSN and GGSN**
   - **SGSN** manages packet data delivery and user mobility.
   - **GGSN** connects UMTS to external networks like the Internet.

Example:

When a user makes a call or uses mobile data, communication flows through **UE → Node B → RNC → Core Network**.

e. Explain the function of a Subscriber Identity Module (SIM) in GSM.

SIM (Subscriber Identity Module) is a smart card used in **GSM (Global System for Mobile Communication)** to identify and authenticate a mobile subscriber.

Functions of SIM

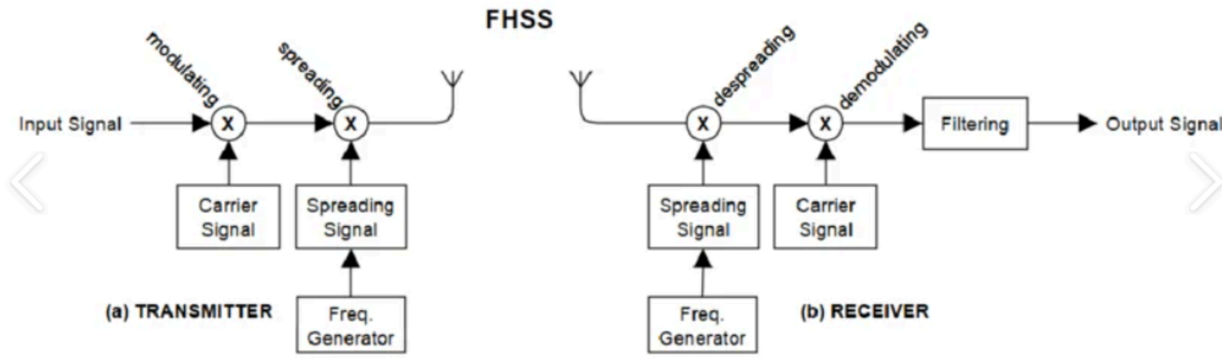
1. **Stores Subscriber Identity:** Contains the **IMSI (International Mobile Subscriber Identity)** that uniquely identifies the user.
2. **Authentication of User:** Verifies the subscriber before allowing access to the GSM network.
3. **Stores Security Keys:** Holds secret keys used for encryption and secure communication.
4. **Enables Network Access:** Allows the user to connect to the mobile network and use services.
5. **Stores User Information:** Can store contacts, SMS messages, and service-related data.
6. **Supports Mobility:** User can move between locations and still access services through the same SIM.
7. **User Portability:** SIM can be inserted into another phone while keeping the same number and subscription.

Q2. [10 Marks each] - Answers

a. What is Frequency Hopping Spread Spectrum (FHSS), and how does it improve communication security?

1. Definition

FHSS is a spread spectrum technique where the carrier frequency rapidly changes (or "hops") among many frequency channels in a pseudo-random sequence known to both transmitter and receiver. Each hop lasts for a short duration before moving to the next frequency.



2. Working Principle

- The available frequency band is divided into multiple narrow channels.
- Transmitter and receiver agree on a pseudo-random hopping sequence.
- Signal hops from one frequency to another at fixed intervals.
- Receiver follows the same hopping pattern to stay synchronized.

3. How FHSS Improves Communication Security

- **Resistance to Jamming:** Since the signal keeps changing frequency, it is difficult for an attacker to jam all channels simultaneously.
- **Low Probability of Interception:** Without knowledge of the hopping sequence, eavesdroppers cannot easily track the communication.
- **Robustness Against Interference:** Narrowband interference affects only a few hops, not the entire transmission.
- **Confidentiality:** The pseudo-random hopping sequence acts like a secret key, adding a layer of security.

b. Explain how LoRaWAN enables long-range, low-power communication by using spread spectrum techniques (Chirp Spread Spectrum).

1. Introduction

LoRaWAN (Long Range Wide Area Network) is a low-power, wide-area networking protocol designed for IoT. It achieves **long-range communication with minimal**

energy consumption by using **LoRa modulation**, which is based on **Chirp Spread Spectrum (CSS)**.

2. Chirp Spread Spectrum (CSS) Technique

- **Chirp Signals:** A chirp is a signal whose frequency increases or decreases over time.
- **Spreading:** CSS spreads the data signal across a wide frequency band using chirps.
- **Orthogonality:** Different spreading factors (SFs) allow multiple devices to transmit simultaneously without interfering.
- **Robustness:** Resistant to noise, multipath fading, and interference.

3. How CSS Enables Long-Range, Low-Power Communication

- **High Sensitivity:** CSS allows receivers to detect signals even below the noise floor, enabling communication over 10–15 km in rural areas.
- **Low Power Consumption:** Devices transmit at low power because CSS ensures reliable decoding even with weak signals.
- **Scalability:** Multiple spreading factors let thousands of devices coexist in the same network.
- **Interference Resistance:** CSS makes signals harder to jam or intercept, improving reliability.
- **Adaptive Data Rate (ADR):** LoRaWAN adjusts spreading factors and transmission rates to balance range, power, and capacity.

Q3. [10 Marks each] - Answers

b. How does Wi-Fi handle multiple users and avoid collisions in the network?

1. Medium Access Control (MAC) in Wi-Fi
Wi-Fi uses **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)**

to manage multiple users on the same wireless medium.

- **Carrier Sense:** Each device listens to the channel before transmitting.
- **Collision Avoidance:** If the channel is busy, the device waits for a random backoff time before retrying.
- **Acknowledgments (ACK):** Receiver sends an ACK after successful packet reception, confirming no collision occurred.
- **Retry Mechanism:** If no ACK is received, the sender retransmits after another backoff.

2. Techniques to Handle Multiple Users

- **Distributed Coordination Function (DCF):** Basic CSMA/CA mechanism for access control.
- **Point Coordination Function (PCF):** Optional polling-based method where the Access Point (AP) schedules transmissions.
- **RTS/CTS (Request to Send / Clear to Send):** Handshake mechanism to reduce hidden node collisions.
- **QoS Enhancements (802.11e):** Prioritizes traffic (voice/video vs. data).
- **OFDMA in Wi-Fi 6 (802.11ax):** Divides channels into subcarriers, allowing multiple users to transmit simultaneously.

Q4. [10 Marks each] - Answers

b. Explain the pairing and security mechanisms used in Bluetooth communication.

1. Pairing in Bluetooth
Pairing is the process of establishing trust between two Bluetooth devices so they can communicate securely.

- **Legacy Pairing (Pre-Bluetooth 2.1):**
  - Devices exchange a PIN code (often 4 digits).

    - The PIN is used to generate a link key for encryption.
    - Vulnerable to eavesdropping and brute-force attacks.
- **Secure Simple Pairing (SSP, Bluetooth 2.1+):**
    - Uses Elliptic Curve Diffie-Hellman (ECDH) for key exchange.
    - Stronger protection against passive and active attacks.
    - Four association models depending on device capabilities:
        - **Just Works:** No user input, but weaker against MITM.
        - **Numeric Comparison:** User confirms matching numbers.
        - **Passkey Entry:** One device displays a code, the other enters it.
        - **Out-of-Band (OOB):** Uses NFC or other channel for key exchange.

2. Security Mechanisms in Bluetooth

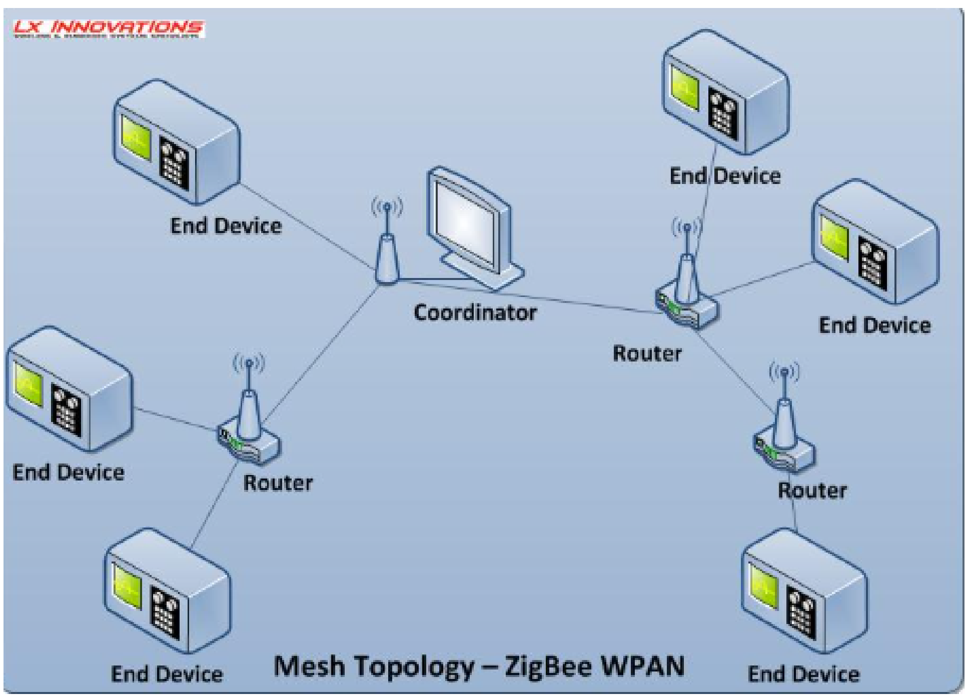
- **Authentication:**
    - Ensures devices are genuine by verifying link keys.
- **Encryption:**
    - Data exchanged is encrypted using stream ciphers (E0 in older versions, AES-CCM in newer).
    - Prevents eavesdropping.
- **Integrity Protection:**
    - Ensures data is not altered during transmission.
- **Frequency Hopping:**
    - Bluetooth uses FHSS (Frequency Hopping Spread Spectrum), making interception harder.
- **Key Management:**
    - Link keys are stored for future connections.
    - Session keys are generated dynamically for each communication.

Q5. [10 Marks each] - Answers

a. What is mesh networking in ZigBee, and how does it enhance the network's reliability?

1. Definition

Mesh networking in ZigBee is a **network topology** where devices (nodes) are interconnected in a way that allows data to travel through multiple paths. Instead of relying on a single central hub, ZigBee mesh networks use **coordinators, routers, and end devices** to forward data across the network.



2. How Mesh Networking Works in ZigBee

- **Coordinator:** Initializes and manages the network.
- **Routers:** Forward data between nodes, creating multiple paths.
- **End Devices:** Simple sensors/actuators that communicate via parent nodes.
- **Multi-hop Communication:** Data can travel across several routers to reach its destination.

- **Self-healing:** If one path fails, the network automatically reroutes data through alternative paths.

3. How Mesh Networking Enhances Reliability

- **Redundancy:** Multiple routes exist between source and destination, reducing the risk of communication failure.
- **Fault Tolerance:** If a node or link fails, data is rerouted through other available nodes.
- **Scalability:** Supports large networks with hundreds of devices.
- **Extended Coverage:** Routers relay data, extending communication range beyond a single hop.
- **Robustness:** Resistant to interference and physical obstacles since data can bypass blocked paths.

b. What are the primary security mechanisms employed in GSM to protect user data and privacy?

GSM (Global System for Mobile Communication) employs several mechanisms to protect **user data and privacy** over the air interface.

1. Authentication

- The network verifies the subscriber using a **challenge-response mechanism**.
- A random number (RAND) is sent to the SIM.
- SIM computes a response (SRES) using the secret key (Ki) and algorithm (A3).
- Prevents unauthorized access to network services.

2. Encryption (Ciphering)

- User data and signaling are encrypted using the **A5 stream cipher**.
- A session key (Kc) is generated from Ki and RAND using algorithm A8.
- Ensures confidentiality of calls and SMS over the air interface.

3. Temporary Identity (TMSI)

- Instead of transmitting the permanent IMSI (International Mobile Subscriber Identity), GSM uses **Temporary Mobile Subscriber Identity (TMSI)**.
- Protects user identity and location privacy from eavesdroppers.

4. Access Control

- Only authenticated and valid subscribers can access network services.
- Prevents fraudulent use of the network.

Q6. [10 Marks each] - Answers

a. Discuss the flaws in WEP's initialization vector (IV) implementation and how it leads to key reuse and susceptibility to attacks.

1. Background
WEP (Wired Equivalent Privacy) was the first security protocol for IEEE 802.11 WLANs. It uses the **RC4 stream cipher** with a secret key and a 24-bit Initialization Vector (IV). The IV is meant to ensure that each packet is encrypted uniquely.

2. Flaws in IV Implementation

- **Small IV Size (24-bit):**
  - Only 224≈16 million possible IVs.
  - In busy networks, IVs repeat quickly (within hours).
- **IV Sent in Cleartext:**
  - IV is transmitted openly in each packet header.
  - Attackers can easily capture IVs without breaking encryption.
- **Weak RC4 Key Scheduling:**
  - WEP combines IV with the secret key to form the RC4 seed.
  - Certain IV values produce weak keys, making it easier to recover the secret key.
- **Key Reuse:**

    - Because IVs repeat, the same RC4 keystream is reused.
    - Attackers can compare ciphertexts encrypted with the same keystream to deduce plaintext.

3. Consequences / Susceptibility to Attacks

- **Statistical Analysis Attacks:** Repeated IVs allow attackers to analyze patterns and recover the key.
- **Known Plaintext Attacks:** If part of the plaintext is known (e.g., headers), attackers can exploit IV reuse to recover keystreams.
- **Key Recovery Attacks (e.g., FMS Attack):** Exploits weak IVs to recover the WEP key in minutes.
- **Practical Breakability:** Tools like *Aircrack-ng* can crack WEP keys by collecting enough packets with reused IVs.

b. How does Cisco implement security in its Unified Wireless Network?

Cisco integrates **multi-layered security mechanisms** into its Unified Wireless Network to protect users, devices, and data.

1. Authentication & Access Control

- **802.1X Authentication:** Uses RADIUS servers for secure user validation.
- **WPA2/WPA3 Support:** Strong encryption and authentication standards.
- **Role-based Access Control (RBAC):** Different policies for employees, guests, and IoT devices.
- **Guest Access Management:** Secure onboarding with temporary credentials.

2. Encryption & Data Protection

- **End-to-End Encryption:** WPA2/WPA3 with AES ensures confidentiality of traffic.

- **Secure Mobility:** Fast Secure Roaming (802.11r) maintains encryption during handoffs.
- **VPN Integration:** Supports secure tunnels for remote users.

3. Intrusion Detection & Prevention

- **Wireless Intrusion Prevention System (wIPS):** Detects rogue APs, unauthorized clients, and denial-of-service attacks.
- **Spectrum Analysis:** Identifies interference sources (microwave, Bluetooth, etc.).
- **Policy Enforcement:** Blocks malicious traffic at the WLAN Controller.

4. Identity & Device Security

- **Cisco Identity Services Engine (ISE):** Provides centralized authentication, authorization, and accounting.
- **Device Profiling:** Identifies and applies policies to smartphones, laptops, IoT devices.
- **MAC Filtering & Certificates:** Adds extra layers of device trust.

5. Management & Monitoring

- **Centralized Control via WLC:** Ensures consistent security policies across all APs.
- **Cisco Prime Infrastructure / DNA Center:** Provides monitoring, compliance checks, and automated alerts.
- **Logging & Auditing:** Tracks user activity for accountability.

Wireless Technology 2023 May PYQ Answers

Q1. [5 Marks each]

a) Explain any four features of MANET and compare MANET and WSN.

b) Write a note on LTE frame structure in detail.

c) Describe the evolution of 1G to 5G mobile systems.

d) Outline the method that supports mobility in CISCO Unified Wireless Network.

e) Write a note on CDMA2000.

Q2. [10 Marks each]

a) Draw and explain UMTS network architecture and compare GSM and UMTS.

b) Draw a neat diagram of GPRS system architecture and explain the function of each block.

Q3. [10 Marks each]

a) Give the significance of WEP protocols. What are the features of WPA2?

b) Draw and explain the architecture of Cisco UWN (Unified Wireless Network) with its features.

Q4. [10 Marks each]

a) Explain the layered architecture of WSN protocol and discuss issues and challenges in WSN.
b) Draw and explain LoRaWAN network architecture and technology stack in detail.

Q5. [10 Marks each]

a) What is Zigbee? Explain the Zigbee protocol stack in detail.

b) Explain 4G network architecture with its specifications.

Q6. [10 Marks each]

a) Describe Bluetooth architecture and protocol stack. Also, discuss its limitations.
b) Draw and explain the protocol architecture of IEEE 802.11. Also, discuss power management in IEEE 802.11 infrastructure networks.

Q1. [5 Marks each] - Answers

a. Explain any four features of MANET and compare MANET and WSN.

(a) Four Features of MANET

MANET (Mobile Ad-hoc Network) is a wireless network formed without fixed infrastructure.

1. **Infrastructure-less Network**
    - No base station or centralized controller is needed.
    - Nodes communicate directly with each other.
2. **Dynamic Topology**
    - Network structure changes frequently because nodes move.
3. **Self-Configuring Network**
    - Nodes automatically join, organize and maintain the network.
4. **Multi-hop Communication**
    - Data is forwarded through intermediate nodes when destination is out of direct range.

(b) Difference between MANET and WSN

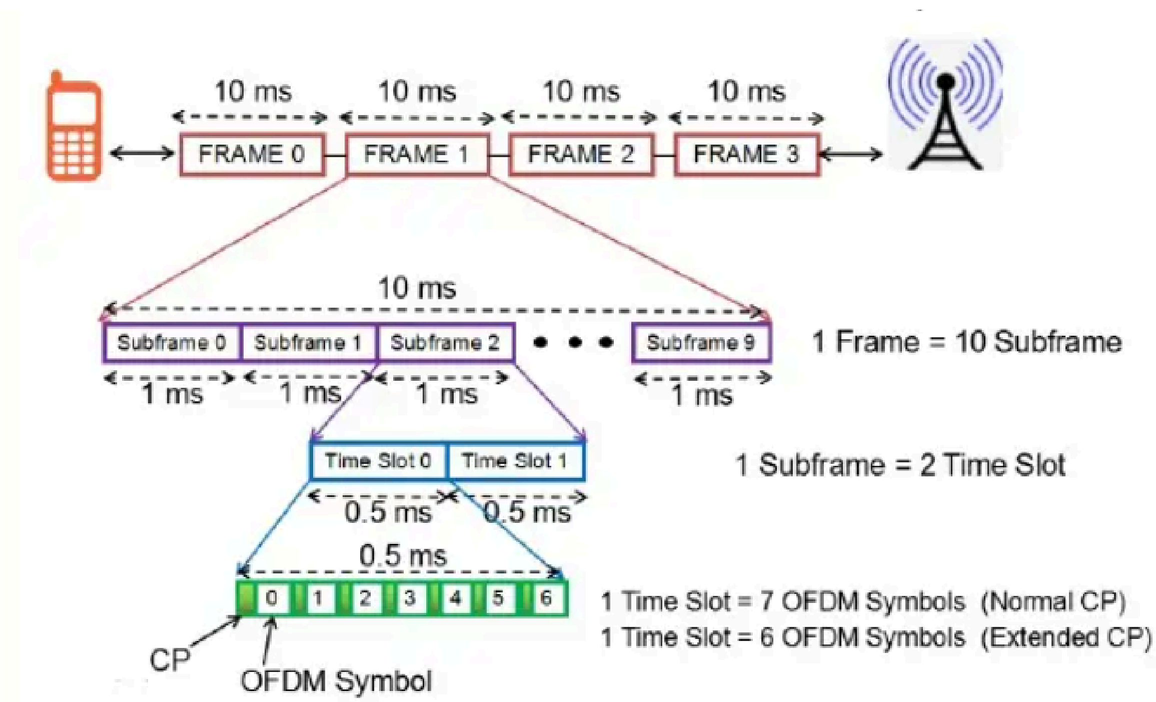
Basis	MANET	WSN
Full Form	Mobile Ad-hoc Network	Wireless Sensor Network

Basis	MANET	WSN
Purpose	General mobile communication	Sensing and monitoring environment
Nodes	Mobile devices like phones, laptops	Sensor nodes
Mobility	Nodes are usually mobile	Nodes are mostly static
Power Source	Comparatively higher power	Limited battery power
Communication	Peer-to-peer communication	Data sent from sensors to sink/base station
Application	Military, disaster recovery	Environmental monitoring, healthcare

b. Write a note on LTE frame structure in detail.

Definition

In LTE (Long Term Evolution), the **frame structure** defines how time and frequency resources are organized for transmission. It ensures efficient multiplexing, synchronization, and scheduling of users.



Key Features of LTE Frame Structure

- **Frame Duration:**
  - Each radio frame = **10 ms**.
  - Divided into **10 subframes**, each of **1 ms**.
- **Slot Structure:**
  - Each subframe consists of **2 slots**, each of **0.5 ms**.
  - Each slot contains a number of OFDM symbols (depends on cyclic prefix).
- **Resource Grid:**
  - Time axis → slots and OFDM symbols.
  - Frequency axis → subcarriers (15 kHz spacing).
  - Together form a **resource block (RB)** = 12 subcarriers × 1 slot.
- **Normal vs Extended Cyclic Prefix:**
  - **Normal CP:** 7 OFDM symbols per slot.
  - **Extended CP:** 6 OFDM symbols per slot (used in large cells to combat delay spread).
- **Downlink Frame:**
  - Based on OFDMA.
  - Control channels (PDCCH, PCFICH, PHICH) occupy first symbols.
  - Data channels (PDSCH) occupy remaining symbols.
- **Uplink Frame:**
  - Based on SC-FDMA (to reduce PAPR).
  - Contains control (PUCCH) and data (PUSCH) channels.
- **Synchronization Signals:**
  - Primary and Secondary Synchronization Signals (PSS, SSS) help UE detect cell identity and frame timing.

c. Describe the evolution of 1G to 5G mobile systems.

1G (First Generation – 1980s)

- **Technology:** Analog voice communication.
- **Access Method:** FDMA.
- **Features:** Poor voice quality, low capacity, no security.
- **Example:** AMPS (Advanced Mobile Phone System).

2G (Second Generation – 1990s)

- **Technology:** Digital voice communication.
- **Access Method:** TDMA/CDMA.
- **Features:** Better voice quality, SMS, basic data services.
- **Example:** GSM, IS-95.

3G (Third Generation – 2000s)

- **Technology:** High-speed data + multimedia.
- **Access Method:** WCDMA/UMTS.
- **Features:** Internet access, video calling, mobile TV.
- **Speed:** Up to a few Mbps.

4G (Fourth Generation – 2010s)

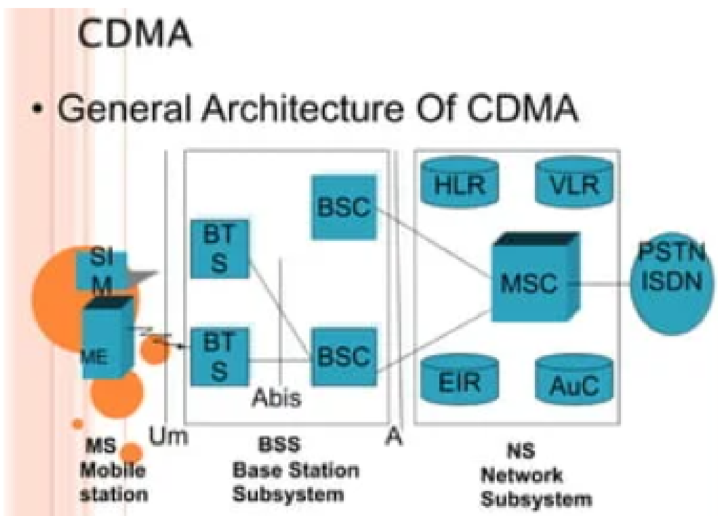
- **Technology:** All-IP packet-switched network.
- **Access Method:** LTE.
- **Features:** High data rates (100 Mbps+), HD streaming, VoIP, gaming, video conferencing.

5G (Fifth Generation – 2020s)

- **Technology:** Ultra-high speed, advanced connectivity.
- **Access Method:** OFDMA, Massive MIMO, mmWave.
- **Features:** Speeds up to 10 Gbps, **low latency (<1 ms)**, massive IoT support, network slicing, enhanced reliability.

e. Write a note on CDMA2000.

CDMA2000 (Code Division Multiple Access 2000) is a **3G mobile communication standard** developed as an evolution of CDMA technology to provide voice and high-speed data services.



Key Features of CDMA2000

1. **Based on CDMA Technology:** Uses spread spectrum and code division techniques for multiple user access.
2. **Supports High Data Rates:** Provides faster data transmission compared to 2G systems.
3. **Voice and Data Services:** Supports voice calls, internet access, multimedia and mobile data.
4. **Efficient Spectrum Usage:** Uses available bandwidth efficiently and supports many users.

5. **Improved Capacity and Coverage:** Offers greater network capacity and better coverage.

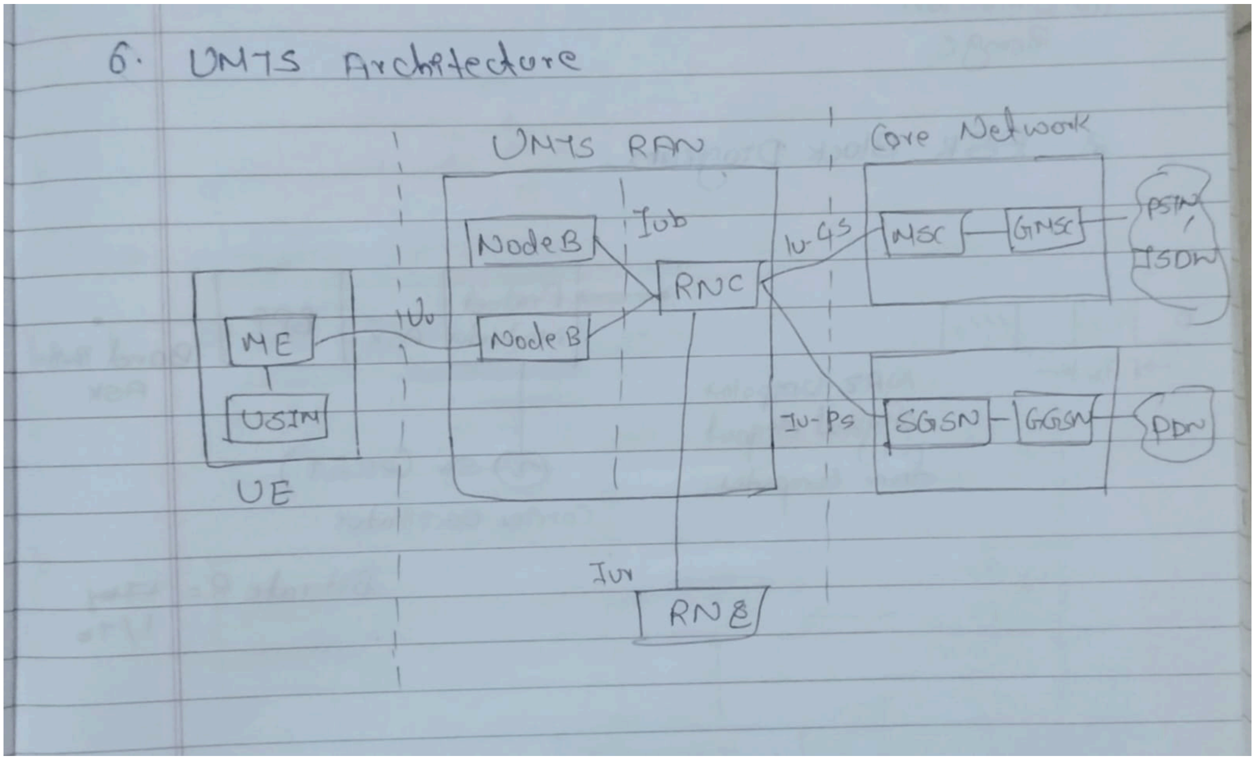
Types of CDMA2000

- **CDMA2000 1x**
  - Supports voice and moderate data rates.
- **CDMA2000 1xEV-DO**
  - Provides high-speed packet data services.
- **CDMA2000 1xEV-DV**
  - Supports both voice and data together.

Q2. [10 Marks each] - Answers

a. Draw and explain UMTS network architecture and compare GSM and UMTS.

UMTS (Universal Mobile Telecommunications System) is a **3G mobile communication system** designed to provide higher data rates and advanced services compared to GSM. Its architecture is divided into three main components:



1. **User Equipment (UE)**
    - Mobile device used by the subscriber.
    - Contains **Mobile Equipment (ME)** and **USIM card** for identity and authentication.
2. **UMTS Terrestrial Radio Access Network (UTRAN)**
    - Provides radio access between UE and the core network.
    - **Node B (Base Stations):** Handles radio transmission/reception.
    - **Radio Network Controller (RNC):** Manages radio resources, handovers, and mobility.
3. **Core Network (CN)**
    - Responsible for switching, routing, and service control.
    - Divided into:
        - **Circuit-Switched Domain:** Voice calls.
        - **Packet-Switched Domain:** Data services (Internet, multimedia).

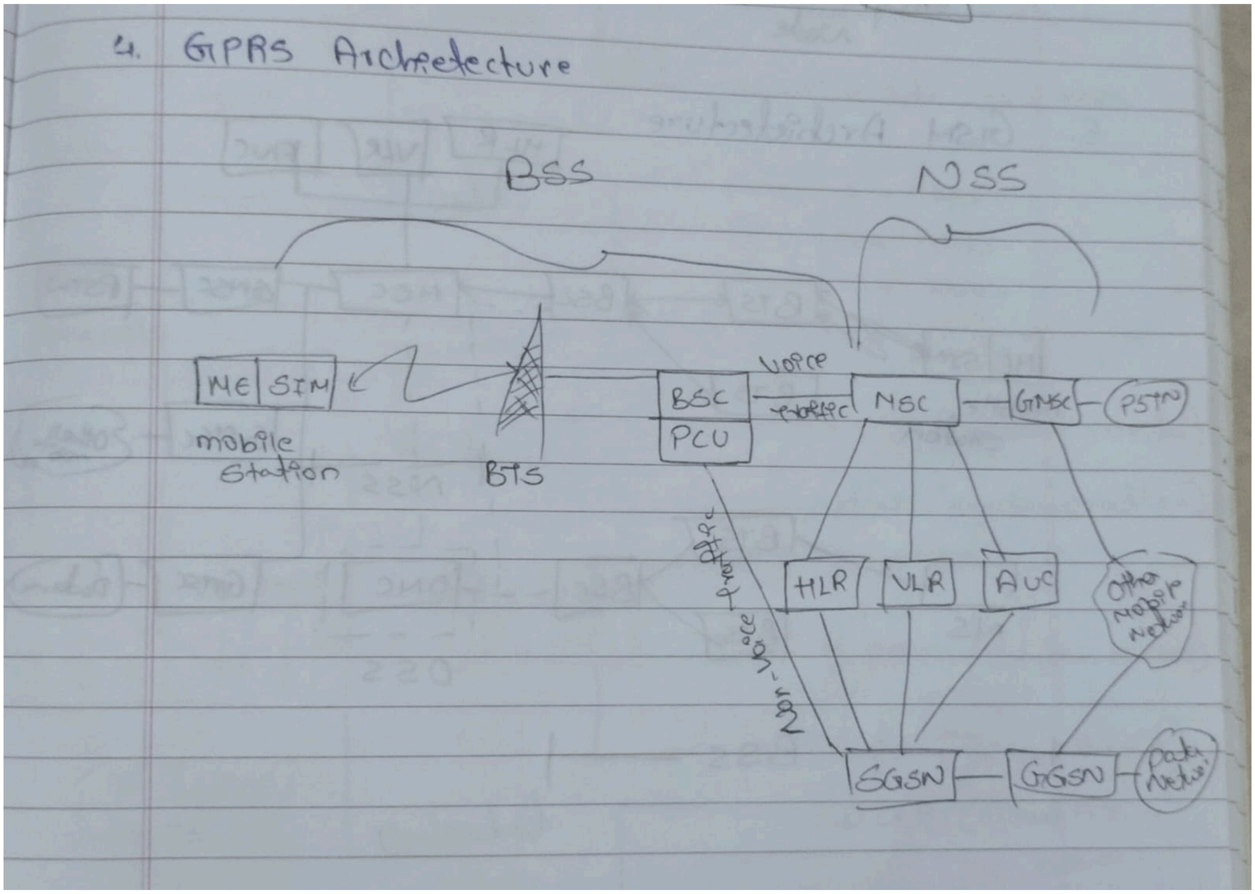
- Interfaces with external networks like PSTN and the Internet.

Feature	GSM (2G)	UMTS (3G)
Architecture	MS + BSS + NSS + OSS	UE + UTRAN + CN
Radio Access	TDMA/FDMA	W-CDMA (Wideband CDMA)
Data Rate	~9.6–14.4 kbps (basic)	Up to 2 Mbps (stationary), 384 kbps (mobile)
Authentication	One-way (network authenticates user)	Mutual authentication (user ↔ network)
Encryption	A5 algorithms (weaker)	Stronger (KASUMI, CK/IK keys)
Services	Voice, SMS, limited data	Voice, video calls, multimedia, Internet
Security	Basic (authentication + encryption)	Enhanced (mutual authentication + integrity protection)
Mobility Management	Location updates via HLR/VLR	Advanced handovers via RNC

b. Draw a neat diagram of GPRS system architecture and explain the function of each block.

1. Introduction

GPRS (General Packet Radio Service) is an enhancement over GSM that enables **packet-switched data services** (e.g., Internet browsing, email, multimedia messaging). Its architecture builds on GSM but adds new nodes for packet handling.



2. Main Components and Functions

- **Mobile Station (MS):**
  - User's handset with GPRS support and SIM card.
  - Initiates and receives packet data services.
- **Base Station Subsystem (BSS):**
  - **BTS (Base Transceiver Station):** Handles radio communication with MS.
  - **BSC (Base Station Controller):** Manages multiple BTSs.
  - **Packet Control Unit (PCU):** Added to BSC for GPRS; directs packet data to SGSN instead of MSC.
- **Serving GPRS Support Node (SGSN):**
  - Core element for packet data.

    - Functions: mobility management, authentication, billing, and routing packets between MS and GGSN.
    - Keeps track of user location and session state.
- **Gateway GPRS Support Node (GGSN):**
    - Connects GPRS network to external packet data networks (e.g., Internet, corporate intranets).
    - Translates GPRS packets into IP packets and vice versa.
    - Allocates IP addresses to mobile stations.
- **Databases (HLR/VLR):**
    - **HLR (Home Location Register):** Stores permanent subscriber data, updated for GPRS services.
    - **VLR (Visitor Location Register):** Temporary data for roaming subscribers.

Q4. [10 Marks each] - Answers

a. Draw and explain LoRaWAN network architecture and technology stack in detail.

LoRaWAN is a **Low-Power Wide-Area Network (LPWAN)** protocol built on **LoRa modulation (Chirp Spread Spectrum)**. Its architecture follows a **star-of-stars topology** with gateways acting as transparent bridges between end devices and the network server.

1. Components of LoRaWAN Architecture

- **End Devices (Nodes):**
    - Battery-powered IoT sensors/actuators.
    - Communicate with gateways using LoRa modulation.
    - Examples: smart meters, soil sensors, trackers.
- **Gateways:**
    - Receive LoRa signals from end devices.

- Forward data via IP (Ethernet, 4G, Wi-Fi) to the network server.
    - Act as transparent bridges, not performing heavy processing.
- **Network Server:**
    - Manages device authentication, routing, and duplicate packet removal.
    - Handles Adaptive Data Rate (ADR) for optimizing power and range.
    - Ensures secure communication with AES-128 encryption.
- **Application Server:**
    - Processes data for end-user applications.
    - Provides dashboards, analytics, and integration with external systems.

Layer	Function
Physical Layer (LoRa)	Chirp Spread Spectrum modulation for long-range, low-power communication
MAC Layer (LoRaWAN)	Defines communication protocol, device classes (A, B, C), ADR, security
Network Layer	Routing, duplicate packet filtering, device management
Application Layer	End-user services (smart city apps, agriculture, industrial IoT)

Q6. [10 Marks each] - Answers

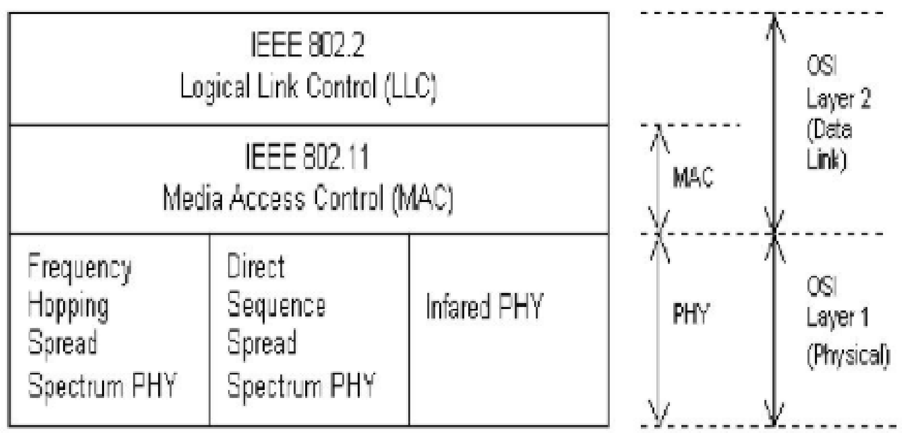
b. Draw and explain the protocol architecture of IEEE 802.11. Also, discuss power management in IEEE 802.11 infrastructure networks.

Definition

IEEE 802.11 is the standard for Wireless LANs (Wi-Fi). Its protocol architecture is layered, ensuring reliable wireless communication between devices.

802.11 protocol architecture

(IEEE 802 protocol layers compared to OSI model)



Layers of IEEE 802.11 Protocol Architecture:

1. **Physical Layer (PHY):**
   - Handles transmission and reception of data over radio waves.
   - Defines modulation techniques (DSSS, OFDM, FHSS).
   - Provides data rates and frequency bands (2.4 GHz, 5 GHz).
2. **MAC Layer (Medium Access Control):**
   - Controls access to the wireless medium.
   - Uses **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance).
   - Provides frame formats, addressing, and retransmission.
   - Supports functions like association, authentication, and encryption.
3. **Logical Link Control (LLC):**
   - Common to all IEEE 802 networks.
   - Provides interface between MAC and higher layers.
   - Ensures error detection and flow control.

4. **Upper Layers (Network & Application):**
    - Standard TCP/IP stack for communication.
    - Applications like web browsing, VoIP, streaming run here.

Power Management in IEEE 802.11 Infrastructure Networks

- **Sleep Mode:** Stations (STAs) can enter low-power state to save battery.
- **Beacon Frames:** Access Point (AP) periodically sends beacons with info about buffered data.
- **Traffic Indication Map (TIM):** Part of beacon frame showing which STAs have pending data.
- **Polling Mechanism:** Sleeping STA wakes up, checks TIM, and requests data.
- **Delivery Traffic Indication Message (DTIM):** Used for multicast/broadcast traffic delivery.
- **Benefit:** Extends battery life while ensuring reliable data transfer.

Wireless Technology 2023 December PYQ Answers

Q1. [5 Marks each]

a) Explain the difference between an Infrastructure-based Network and an Ad-hoc Network of WLAN.
b) Explain Direct Sequence Spread Spectrum (DSSS).
c) Compare FDMA, TDMA, and CDMA.
d) Outline the method that supports mobility in CISCO Unified Wireless Network.
e) State the features of Wi-MAX.

Q2. [10 Marks each]

a) Draw and explain 4G architecture with its specifications.
b) Explain in detail GSM Architecture with a neat diagram.

Q3. [10 Marks each]

a) Give the significance of WEP protocols. What are the features of WPA2?
b) Draw and explain Cisco UWN lightweight AP and WLC Operation.

Q4. [10 Marks each]

a) What are the various challenges in WSN and explain the architecture of WSN?
b) What do you mean by massive MIMO? Compare 1G-5G mobile standards.

Q5. [10 Marks each]

a) Explain Zigbee protocol stack.
b) Give the Features of VANET, E-VANET, and MANET.

Q6. [10 Marks each]

a) Explain UMTS and GSM security.
b) Draw and explain IEEE 802.11 Architecture.

Q1. [5 Marks each] - Answers

c. Compare FDMA, TDMA, and CDMA.

Basis	FDMA	TDMA	CDMA
Full Form	Frequency Division Multiple Access	Time Division Multiple Access	Code Division Multiple Access
Access Method	Users are separated by different frequencies	Users share same frequency using time slots	Users share same frequency using unique codes
Channel Division	Divided by frequency bands	Divided by time slots	Divided by codes
Bandwidth Usage	Less efficient	More efficient than FDMA	Highly efficient
Interference	More adjacent channel interference	Less interference than FDMA	Very low interference due to spread spectrum
Capacity	Lower user capacity	Higher than FDMA	Highest capacity
Synchronization	Less synchronization needed	Requires time synchronization	Requires code synchronization
Security	Low security	Moderate security	High security
Example	1G Analog systems	GSM	CDMA-based 3G systems